Four results on the critical 2D KPZ process

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Abstract

We state and prove, or sketch the proofs of, four properties of the critical two-dimensional KPZ process constructed by Berger, Caravenna, Clark, Dunlap, Gu, Liu, Nakajima, Sun, Tsai, Xie and Zygouras. Averages of the flat field against nonnegative test functions have strictly positive third central moment; the proof combines the representation of their laws as nondecreasing convex functions of a Gaussian variable with a cubic inequality. The field depends twice differentiably on the initial condition, and the two derivatives are limits of averaged means and variances of the initial perturbation under tilted polymer measures. The averages of the flat field at the scales $\exp(-\mathrm{e}^{Ls})$, divided by $\sqrt{L}$, converge as continuous processes in s to a Brownian motion, because nested scales see nested sums of independent time blocks of the stochastic heat flow. Finally, after the common fluctuation is removed by testing against functions of zero integral, the field seen at scale ε and multiplied by $\sqrt{\log(1/\varepsilon)}$ converges in finite-dimensional distributions to a Gaussian free field; the main new estimate shows that logarithms of positively associated masses inherit the spatial decay of their covariance. All four theorems have been formally verified in Lean 4.

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1 Introduction

Berger et al. [BCC+26] construct the critical two-dimensional KPZ process $\mathscr{H}^\vartheta = (\mathscr{H}_t^\vartheta(h_0,\cdot))_{t>0}$ as a renormalised logarithm of the critical two-dimensional stochastic heat flow $\mathscr{Z}^\vartheta$ of Caravenna, Sun and Zygouras [CSZ23], where $\vartheta \in \mathbb{R}$ fixes the position in the critical window. For each time $t > 0$ and bounded continuous initial condition h_0 it is a random distribution on $\mathbb{R}^2$, and all these distributions are built from one realisation of the stochastic heat flow. They also identify its small-scale structure: averages of the field over a ball of radius ε are Gaussian at the scale $\sqrt{\log\log(1/\varepsilon)}$, and the covariance kernel of the field blows up like $\log\log(1/|z|)$ at the diagonal [BCC+26, Theorem 1.4]; yet no average of the field against a nonzero nonnegative test function is Gaussian [BCC+26, Theorem 1.6]. In this note we prove four further properties of the field. Section 2 recalls the construction and the results of [BCC+26] that we use. We write $\mathscr{H}_t^\vartheta(h_0,\psi)$ for the action of $\mathscr{H}_t^\vartheta(h_0,\cdot)$ on a test function $\psi \in C_c^\infty(\mathbb{R}^2)$, and $\mathscr{H}_t^\vartheta(\psi) = \mathscr{H}_t^\vartheta(\mathbf{0},\psi)$ for the flat initial condition. For a function φ on $\mathbb{R}^2$, $\varepsilon > 0$ and $x \in \mathbb{R}^2$ we put

$$\varphi_\varepsilon^{(x)}(y) = \varepsilon^{-2}\varphi\big(\varepsilon^{-1}(y-x)\big)\,. \tag{1.1}$$

The first result strengthens the non-Gaussianity theorem of [BCC+26] to a sign: averages of the field against nonnegative test functions are skewed to the right.

Theorem A (Strict skewness)**.** *Let $t > 0$, $\vartheta \in \mathbb{R}$, and let $\psi \in C_c^\infty(\mathbb{R}^2)$ be nonnegative and not identically zero. Then $X = \mathscr{H}_t^\vartheta(\psi)$ satisfies $\mathbb{E}|X|^3 < \infty$ and*

$$\mathbb{E}\big[(X - \mathbb{E}X)^3\big] > 0\,.$$

The second result concerns the dependence of the field on its initial condition. Write $g_r(z) = (2\pi r)^{-1}\mathrm{e}^{-|z|^2/(2r)}$ for the heat kernel, $g_r^{(x)} = g_r(\cdot - x)$, and $\mathscr{Z}_{s,t}^\vartheta(f,f') = \iint f(y)\,f'(y')\,\mathscr{Z}_{s,t}^\vartheta(\mathrm{d}y,\mathrm{d}y')$ for the stochastic heat flow tested against nonnegative functions: f weighs the position of the polymer at time s and f' its position at time t; we also write $\mathscr{Z}_{s,t}^\vartheta(\mathrm{d}y,f') = \int f'(y')\,\mathscr{Z}_{s,t}^\vartheta(\mathrm{d}y,\mathrm{d}y')$. For $t > 0$, $\vartheta \in \mathbb{R}$, bounded continuous h, v, a test function ψ, a smoothing parameter $r > 0$ and $q \in \mathbb{R}$, define

$$\mathcal{R}_r(q) = \int_{\mathbb{R}^2} \psi(x)\Big\{\log \mathscr{Z}_{0,t}^\vartheta\big(\mathrm{e}^{h+qv}, g_r^{(x)}\big) - \log \mathscr{Z}_{0,t}^\vartheta\big(\mathrm{e}^{h}, g_r^{(x)}\big)\Big\}\,\mathrm{d}x\,. \tag{1.2}$$

Unlike the regularisation (2.1) used to construct the field, $\mathcal{R}_r$ keeps the disorder of the whole interval $[0,t]$ and smooths only the position at time t; no centring is needed, because it would cancel in the difference. The logarithms are almost surely finite and integrable against ψ (Section 4). We give $C^2(\mathbb{R})$ the topology of uniform convergence on compact sets of functions and their first two derivatives.

Theorem B (Response to the initial condition)**.** *Let $t > 0$, $\vartheta \in \mathbb{R}$, let $h, v : \mathbb{R}^2 \to \mathbb{R}$ be bounded and continuous, and let $\psi \in C_c^\infty(\mathbb{R}^2)$. Put $A = \|\psi\|_{L^1(\mathbb{R}^2)}$ and $M = \|v\|_{L^\infty(\mathbb{R}^2)}$. There exists a random variable $\mathcal{R}$ with values in $C^2(\mathbb{R})$, unique up to almost sure equality, with the following properties.*

(i) For every $q \in \mathbb{R}$, almost surely, $\mathcal{R}(q) = \mathscr{H}_t^\vartheta(h+qv,\psi) - \mathscr{H}_t^\vartheta(h,\psi)$.

(ii) Almost surely, for every $q \in \mathbb{R}$,

$$|\mathcal{R}(q)| \le AM|q|\,, \qquad |\mathcal{R}'(q)| \le AM\,, \qquad |\mathcal{R}''(q)| \le AM^2\,,$$

and $\mathcal{R}''(q) \ge 0$ if $\psi \ge 0$.

(iii) For every $r > 0$, almost surely $\mathcal{R}_r \in C^3(\mathbb{R})$; and for all $a \le b$ and $p \in [1, \infty)$,

$$\lim_{r \downarrow 0} \mathbb{E}\Big[\sup_{q \in [a,b]} \max_{j \in \{0,1,2\}} \big| \mathcal{R}_r^{(j)}(q) - \mathcal{R}^{(j)}(q) \big|^p \Big] = 0 \,.$$

For positive r, Proposition 4.1 shows that

$$\mathcal{R}_r'(q) = \int_{\mathbb{R}^2} \psi(x)\, \pi_{r,q}^x(v)\, \mathrm{d}x \,, \qquad \mathcal{R}_r''(q) = \int_{\mathbb{R}^2} \psi(x)\, \mathrm{Var}_{\pi_{r,q}^x}(v)\, \mathrm{d}x \,,$$

where $\pi_{r,q}^x(\mathrm{d}y) \propto \mathrm{e}^{h(y)+qv(y)}\, \mathscr{Z}_{0,t}^{\vartheta}(\mathrm{d}y, g_r^{(x)})$ is the law of the position at time 0 of a polymer that ends near x at time t, tilted by e^{h+qv}. Theorem B thus identifies $\mathcal{R}'$ and $\mathcal{R}''$ as limits of ψ-weighted integrals of the means and variances of v under these laws.

The last two results describe the field at small scales. Theorem 1.4 of [BCC+26] shows that $\mathscr{H}_t^{\vartheta}(\varphi_\varepsilon^{(x)})/\sqrt{\log\log(1/\varepsilon)}$ is asymptotically Gaussian for nonnegative φ. Writing $\varepsilon = \exp(-\mathrm{e}^{Ls})$ turns $\log\log(1/\varepsilon)$ into Ls, and the next theorem shows that, along this change of variables, the averages behave jointly in s like a Brownian motion.

Theorem C (Brownian motion across scales)**.** *Let $t > 0$, $\vartheta \in \mathbb{R}$, $x \in \mathbb{R}^2$ and $S > 0$, and let $\varphi \in C_c^\infty(\mathbb{R}^2)$ be nonnegative with $\int \varphi = 1$. For $L > 0$ and $s \in [0, S]$ put*

$$X_L(s) = \frac{1}{\sqrt{L}}\, \mathscr{H}_t^{\vartheta}\big(\varphi_{\varepsilon_L(s)}^{(x)}\big) \,, \qquad \varepsilon_L(s) = \exp\big(-\mathrm{e}^{Ls}\big) \,.$$

Then every path $s \mapsto X_L(s)$ is continuous on $[0, S]$, so that X_L is a random variable in $C([0, S])$, and as $L \to \infty$ the processes $(X_L(s))_{s \in [0,S]}$ converge in distribution in $C([0, S])$, with the supremum norm, to a standard Brownian motion $(B_s)_{s \in [0,S]}$ started at 0.

Averages against functions of zero integral cancel the large, nearly constant part of the field that drives Theorem 1.4 of [BCC+26] and Theorem C. What remains is much smaller, and the last result identifies it as a Gaussian free field.

Theorem D (Gaussian free field tangent)**.** *Let $t > 0$, $\vartheta \in \mathbb{R}$ and $x \in \mathbb{R}^2$, and let $f_1, \dots, f_k \in C_c^\infty(\mathbb{R}^2)$ be real with $\int f_i = 0$ for every i. Then, as $\varepsilon \downarrow 0$,*

$$\Big(\sqrt{\log(1/\varepsilon)}\, \mathscr{H}_t^{\vartheta}\big(f_{i,\varepsilon}^{(x)}\big) \Big)_{i=1}^k \implies \mathcal{N}(0, \mathsf{Q}) \,, \qquad \mathsf{Q}_{ij} = -\iint_{\mathbb{R}^2 \times \mathbb{R}^2} f_i(u)\, f_j(u')\, \log|u - u'|\, \mathrm{d}u\, \mathrm{d}u' \,,$$

where $f_{i,\varepsilon}^{(x)}$ is defined by (1.1).

The integrals defining Q converge absolutely, and Q is positive semidefinite (Lemma A.1), so the Gaussian law $\mathcal{N}(0, \mathsf{Q})$ exists; it may be degenerate. Since $-\log|z| = 2\pi G(z)$, where G is the Green function of $-\Delta$ on $\mathbb{R}^2$, $\mathcal{N}(0, \mathsf{Q})$ is the law of $(\Psi(f_i))_{i \le k}$ for the whole-plane Gaussian free field Ψ with covariance kernel $-\log|u - u'|$, which is defined on test functions of zero integral. In this sense Ψ is a tangent field of $\mathscr{H}_t^{\vartheta}$ at x; its law does not depend on t, ϑ or x.

Ideas of the proofs. Theorem A rests on a representation from [BCC+26]: by Gaussian convexity, the law of $\mathscr{H}_t^{\vartheta}(\psi)$ is that of $F(G)$, with G a standard Gaussian variable and F nondecreasing and convex. We show that for such F the third central moment of $F(G)$ bounds the cubic distance between $F(G)$ and the Gaussian variable with the same mean and variance coupled to it through G;

the non-Gaussianity theorem of [BCC+26] then makes the third moment positive. For Theorem B, at a positive smoothing parameter the response is a logarithmic moment generating function, so its derivatives are cumulants of the bounded perturbation, bounded uniformly in the smoothing; the result of [BCC+26] that smoothing the position at time t yields the same field gives the limit at each fixed q, and finite differences turn this into convergence of two derivatives. The finite-dimensional part of Theorem C comes from independent time blocks of the stochastic heat flow, which make the averages at nested scales close to nested partial sums of one independent sequence, and its tightness from spatial stationarity, through a truncated stationary maximal inequality applied to a band-limited version of the field. For Theorem D, scaling turns the small spatial scale into a weakly disordered flow run over a critically long time. The late part of that time interval cancels against tests of zero integral, the early part is close to a Gaussian linear functional, and the passage from a positive smoothing scale to the field itself uses a new estimate: logarithms of positively associated masses inherit the spatial decay of the covariance of the masses, up to factors polynomial in their moments.

Scope. The proofs of Theorems A and B are complete, given the results of [BCC+26] that they quote. For Theorems C and D we combine complete arguments with proof sketches that indicate the estimates used; the sketches include the spectral remainder estimates of Section 5 and the improved endpoint-profile estimate of Lemma 6.3. The openings of Sections 5 and 6 specify which parts are proved completely. All four theorems, including the parts whose proofs are only sketched here, have been formally verified in Lean 4; the formalization is available at `https://github.com/scottnarmstrong/Critical2DKPZ`. We have not searched the literature systematically, and we make no claim of priority beyond the classical antecedent noted in Section 3.

Organisation. Section 2 recalls the setting. Sections 3, 4, 5 and 6 prove Theorems A, B, C and D, and Appendix A collects facts about the logarithmic kernel.

2 Setting

We use the objects constructed in [BCC+26] and quote its results by number. This section recalls the results used in more than one of the later sections; a result used in one argument only is quoted where it is used.

The stochastic heat flow. The critical two-dimensional stochastic heat flow is a family $\mathscr{Z}^\vartheta = (\mathscr{Z}^\vartheta_{s,t}(\mathrm{d}y, \mathrm{d}y'))_{0\le s<t}$ of random locally finite measures on $\mathbb{R}^2 \times \mathbb{R}^2$. It is obtained by Caravenna, Sun and Zygouras as the scaling limit of directed polymer partition functions in the critical window [CSZ23, Theorems 1.1 and 9.1], see [BCC+26, Theorem A]. Its mean is the heat kernel: $\mathbb{E}[\mathscr{Z}^\vartheta_{s,t}(\mathrm{d}y, \mathrm{d}y')] = g_{t-s}(y'-y)\,\mathrm{d}y\,\mathrm{d}y'$. We use the notation g_r, $g_r^{(x)}$ and $\mathscr{Z}^\vartheta_{s,t}(f, f')$ of Section 1.

The height field. For bounded continuous $h_0 : \mathbb{R}^2 \to \mathbb{R}$ and $0 < \delta < t$, the regularised height of [BCC+26, (1.18)] removes the disorder of the time interval $[t-\delta, t]$:

$$\mathscr{H}_t^{\vartheta,(\delta)}(h_0, x) = \log \mathscr{Z}^\vartheta_{0,t-\delta}\big(\mathrm{e}^{h_0}, g_\delta^{(x)}\big) - c_t^{\vartheta,(\delta)}\,, \qquad c_t^{\vartheta,(\delta)} = \mathbb{E}\big[\log \mathscr{Z}^\vartheta_{0,t-\delta}\big(\mathbf{1}, g_\delta^{(x)}\big)\big]\,. \tag{2.1}$$

The centring is computed for the flat initial condition, and by translation invariance it does not depend on x. Theorem 1.2 of [BCC+26] states that there is a process of random distributions

$\mathscr{H}_t^{\vartheta}(h_0,\cdot)$, a measurable function of the stochastic heat flow, such that, jointly for finitely many $t>0$ and h_0, and for every $a>0$,

$$\mathscr{H}_t^{\vartheta,(\delta)}(h_0,\cdot) \xrightarrow[\delta\downarrow 0]{\mathbb{P}} \mathscr{H}_t^{\vartheta}(h_0,\cdot) \qquad \text{in } H^{-a}_{\mathrm{loc}}(\mathbb{R}^2)\,.$$

The same theorem states that for each $\psi\in C_c^\infty(\mathbb{R}^2)$ the tested fields converge in $L^p(\mathbb{P})$ for every $p\in[1,\infty)$, so that

$$\mathbb{E}\big|\mathscr{H}_t^{\vartheta}(h_0,\psi)\big|^p<\infty \qquad \text{for every } p\in[1,\infty)\,. \tag{2.2}$$

All the fields are built from the same stochastic heat flow and the same centring. This joint construction is what makes the difference $\mathscr{H}_t^{\vartheta}(h+qv,\psi)-\mathscr{H}_t^{\vartheta}(h,\psi)$ in Theorem B meaningful. Sections 3, 5 and 6 concern only the flat field $\mathscr{H}_t^{\vartheta}(\psi)=\mathscr{H}_t^{\vartheta}(\mathbf{0},\psi)$. Our results depend only on the joint law of the flow and the fields. They therefore hold for any random distributions defined on the probability space of a critical stochastic heat flow and obtained as limits in probability of the regularised heights (2.1), since such limits are almost surely unique.

Symmetries. By [BCC+26, Proposition 1.8], the flat field is invariant in law under rotations and translations, and it has the following scaling covariance: for every $\kappa>0$, jointly for finitely many times t_i and test functions φ_i,

$$\big(\mathscr{H}^{\vartheta}_{\kappa t_i}(\mathcal{S}_\kappa\varphi_i)\big)_i \overset{\mathrm{d}}{=} \big(\mathscr{H}^{\vartheta+\log\kappa}_{t_i}(\varphi_i)\big)_i\,, \qquad \mathcal{S}_\kappa\varphi(y)=\kappa^{-1}\varphi\big(\kappa^{-1/2}y\big)\,. \tag{2.3}$$

Since $\varphi_\varepsilon^{(0)}=\mathcal{S}_{\varepsilon^2}\varphi$, the field at time t seen at scale ε has the law of the field at time $t\varepsilon^{-2}$, at parameter $\vartheta-2\log(1/\varepsilon)$, seen at scale one. Sections 5 and 6 use this reduction.

Local structure. Theorem 1.4 of [BCC+26] has two parts. First, for nonnegative $\varphi\in C_c^\infty(\mathbb{R}^2)$ and $x\in\mathbb{R}^2$,

$$\frac{\mathscr{H}_t^{\vartheta}(\varphi_\varepsilon^{(x)})}{\sqrt{\log\log(1/\varepsilon)}} \implies \mathcal{N}\Big(0,\Big(\int\varphi\Big)^2\Big) \qquad \text{as } \varepsilon\downarrow 0\,. \tag{2.4}$$

Second, there is a nonnegative radial kernel $\mathsf{C}_t^{\vartheta}\in L^2_{\mathrm{loc}}(\mathbb{R}^2)$, continuous off the origin, such that for real $f,f'\in C_c^\infty(\mathbb{R}^2)$

$$\mathrm{Cov}\big(\mathscr{H}_t^{\vartheta}(f),\mathscr{H}_t^{\vartheta}(f')\big)=\iint f(y)\,f'(y')\,\mathsf{C}_t^{\vartheta}(y-y')\,\mathrm{d}y\,\mathrm{d}y'\,, \qquad \lim_{|z|\downarrow 0}\frac{\mathsf{C}_t^{\vartheta}(z)}{\log\log(1/|z|)}=1\,. \tag{2.5}$$

Theorem C refines (2.4) jointly across scales; Theorem D concerns the next order of C_t^{ϑ}, which (2.5) does not see.

3 Strict skewness

This section proves Theorem A, which sharpens [BCC+26, Theorem 1.6], the non-Gaussianity of $\mathscr{H}_t^{\vartheta}(\psi)$ for nonnegative ψ, to a sign. Both rest on the Gaussian convexity of the model. The logarithm of a polymer partition function is a convex, nondecreasing function of the Gaussian disorder; by a consequence of Ehrhard's inequality such a function has the law of $F(G)$, with G a standard Gaussian variable and F convex and nondecreasing, and this representation survives in the limit (Section 3.2). A nondecreasing convex function stretches the upper tail of G and compresses the lower one, so $F(G)$ should be skewed to the right. We prove this in the quantitative

form of Proposition 3.1: the third central moment of $F(G)$ controls $\mathbb{E}|F(G) - m - \sigma G|^3$, the cubic distance from $F(G)$ to the Gaussian variable $m + \sigma G$ with the same mean m and variance σ^2. Equality holds only for affine F, which non-Gaussianity rules out.

3.1 Convex functions of a Gaussian variable

Throughout this subsection G is a standard Gaussian variable and γ its law on $\mathbb{R}$.

Proposition 3.1. *Let $F : \mathbb{R} \to \mathbb{R}$ be nondecreasing and convex with $\mathbb{E}|F(G)|^3 < \infty$. Put $m = \mathbb{E}F(G)$ and $\sigma = \mathrm{Var}(F(G))^{1/2}$. Then*

$$\mathbb{E}\big|F(G) - m - \sigma G\big|^3 \le 2\,\mathbb{E}\big[(F(G) - m)^3\big]\,. \tag{3.1}$$

In particular $\mathbb{E}[(F(G) - m)^3] \ge 0$, with equality if and only if $F(x) = \alpha x + \beta$ for some $\alpha \ge 0$ and $\beta \in \mathbb{R}$.

The inequality $\mathbb{E}[(F(G) - m)^3] \ge 0$ is classical: van Zwet showed that nondecreasing convex transformations do not decrease standardised odd moments [vZ64, Theorem 2.2.1], by counting the crossings of the transformation with an affine function. What we need beyond his weak inequality is the characterisation of equality for an arbitrary nondecreasing convex F with a finite third moment, without regularity or exponential integrability; the bound (3.1) provides it.

The proof compares the centred function $F - m$ with the line $x \mapsto \sigma x$, which has the same mean and the same second moment under γ. Matching these two moments forces $F - m$ to cross the line exactly twice (Lemma 3.2, a version of van Zwet's crossing argument, cf. [vZ64, p. 10]). A cubic polynomial with critical points at the two crossing values then separates the third moments of the two functions (Lemma 3.3): its linear and quadratic terms are invisible to the matched moments.

Lemma 3.2. *Let F, m and σ be as in Proposition 3.1, and assume $\sigma > 0$. Put $\bar F = F - m$ and $\rho(x) = \bar F(x) - \sigma x$. If ρ is not identically zero, then there are real numbers $x_1 < x_2$ such that $\rho < 0$ on (x_1, x_2), $\rho \ge 0$ on $\mathbb{R} \setminus (x_1, x_2)$, and $\rho(x_1) = \rho(x_2) = 0$.*

Proof. A finite convex function on $\mathbb{R}$ is continuous, so ρ is continuous; it is convex as the difference of a convex and a linear function. The function $w(x) = \bar F(x) + \sigma x$ satisfies $\rho w = \bar F^2 - \sigma^2 x^2$, so it encodes the matching of second moments, and it is strictly increasing, since $\bar F$ is nondecreasing and $\sigma > 0$. Both $\rho(G)$ and $w(G)$ are square integrable, and because $\bar F(G)$ and σG have the same mean and the same second moment,

$$\mathbb{E}[\rho(G)] = 0\,, \qquad \mathbb{E}[\rho(G)\,w(G)] = \mathbb{E}\big[\bar F(G)^2\big] - \sigma^2\mathbb{E}[G^2] = 0\,. \tag{3.2}$$

Step 1: ρ changes sign. If $\rho \ge 0$ everywhere, the first identity in (3.2) gives $\rho(G) = 0$ almost surely; since ρ is continuous and γ charges every open interval, ρ vanishes identically. Hence ρ takes negative values, and in the same way positive ones. The set $\{\rho < 0\}$ is open by continuity and an interval by convexity, so it equals (x_1, x_2) with $-\infty \le x_1 < x_2 \le \infty$, and it is not all of $\mathbb{R}$.

Step 2: neither endpoint is infinite. Suppose $x_1 = -\infty$, so that x_2 is finite and $\rho(x_2) = 0$. For $x < x_2$ both $\rho(x)$ and $w(x) - w(x_2)$ are negative, and for $x \ge x_2$ both are nonnegative. Therefore $\rho(x)(w(x) - w(x_2)) \ge 0$ for all x, with strict inequality on $(-\infty, x_2)$, which γ charges, so by (3.2)

$$0 < \mathbb{E}\big[\rho(G)\big(w(G) - w(x_2)\big)\big] = \mathbb{E}[\rho(G)\,w(G)] - w(x_2)\,\mathbb{E}[\rho(G)] = 0\,,$$

a contradiction. If $x_2 = +\infty$, the same product with $w(x_1)$ in place of $w(x_2)$ is nonpositive everywhere and negative on (x_1, ∞), which contradicts (3.2) in the same way. Thus x_1 and x_2 are finite, $\rho \ge 0$ off (x_1, x_2) by definition, and $\rho(x_1) = \rho(x_2) = 0$ by continuity. □

For real numbers $a_1 < a_2$, the cubic $P(y) = y^3 - \frac{3}{2}(a_1 + a_2)y^2 + 3a_1a_2\,y$ has $P'(y) = 3(y - a_1)(y - a_2)$, so it increases on $(-\infty, a_1]$ and on $[a_2, \infty)$ and decreases on $[a_1, a_2]$. In each case of the next lemma, y and z lie in the same one of these intervals and $P(y) \geq P(z)$; the lemma quantifies this inequality.

Lemma 3.3. *Let $a_1 < a_2$ and $P(y) = y^3 - \frac{3}{2}(a_1 + a_2)y^2 + 3a_1a_2\,y$. Let $y, z \in \mathbb{R}$ satisfy one of*

$$\textit{(a) } z \leq y \leq a_1\,, \qquad \textit{(b) } a_1 \leq y \leq z \leq a_2\,, \qquad \textit{(c) } a_2 \leq z \leq y\,.$$

Then $P(y) - P(z) \geq \frac{1}{2}|y - z|^3$.

Proof. Each case follows from a polynomial identity whose right side is a sum of products of nonnegative factors:

$$\begin{aligned}
&\text{(a)} \quad P(y) - P(z) - (y - z)^3 = \tfrac{3}{2}(y - z)^2\big((a_1 - y) + (a_2 - y)\big) + 3(y - z)(a_1 - y)(a_2 - y)\,,\\
&\text{(b)} \quad 2\big(P(y) - P(z)\big) - (z - y)^3 = 3(z - y)^2\big((y - a_1) + (a_2 - z)\big) + 6(z - y)(y - a_1)(a_2 - z)\,,\\
&\text{(c)} \quad P(y) - P(z) - (y - z)^3 = \tfrac{3}{2}(y - z)^2\big((z - a_1) + (z - a_2)\big) + 3(y - z)(z - a_1)(z - a_2)\,.
\end{aligned}$$

Case (b) determines the constant $\frac{1}{2}$; cases (a) and (c) even give $P(y) - P(z) \geq |y - z|^3$. □

Proof of Proposition 3.1. Since $\mathbb{E}|F(G)|^3 < \infty$, every polynomial of degree at most three in $F(G)$ and G is integrable, and all expectations below are finite. We use the notation $\bar{F} = F - m$ and $\rho(x) = \bar{F}(x) - \sigma x$ of Lemma 3.2. If $\sigma = 0$, then $\bar{F}(G) = 0$ almost surely, so $\bar{F}$ vanishes identically by continuity and full support of γ: F is constant, both sides of (3.1) vanish, and F is affine with $\alpha = 0$. If $\sigma > 0$ and ρ vanishes identically, then $F(x) = m + \sigma x$, and both sides of (3.1) vanish because $\mathbb{E}G^3 = 0$. These two cases are exactly the affine F: if $F(x) = \alpha x + \beta$ with $\alpha \geq 0$, then $\sigma = \alpha$, and $\rho \equiv 0$ when $\alpha > 0$.

Assume now that $\sigma > 0$ and ρ is not identically zero, and let $x_1 < x_2$ be given by Lemma 3.2. Put $a_i = \sigma x_i$, so that $a_1 < a_2$ and $\bar{F}(x_i) = a_i$. The sign of ρ and the monotonicity of $\bar{F}$ and of $x \mapsto \sigma x$ give, for every $x \in \mathbb{R}$,

$$\begin{aligned}
\sigma x \leq \bar{F}(x) \leq a_1 \qquad &\text{if } x \leq x_1\,,\\
a_1 \leq \bar{F}(x) \leq \sigma x \leq a_2 \qquad &\text{if } x_1 \leq x \leq x_2\,,\\
a_2 \leq \sigma x \leq \bar{F}(x) \qquad &\text{if } x \geq x_2\,.
\end{aligned}$$

These are the three configurations of Lemma 3.3 with $y = \bar{F}(x)$ and $z = \sigma x$. With P as in that lemma,

$$P\big(\bar{F}(x)\big) - P(\sigma x) \geq \tfrac{1}{2}\big|\bar{F}(x) - \sigma x\big|^3 \qquad \text{for every } x \in \mathbb{R}\,. \tag{3.3}$$

We evaluate (3.3) at $x = G$ and take expectations. Since $\bar{F}(G)$ and σG have the same first and second moments, the quadratic and linear terms of P cancel, and $\mathbb{E}[(\sigma G)^3] = 0$; hence

$$\mathbb{E}\big[P(\bar{F}(G)) - P(\sigma G)\big] = \mathbb{E}\big[\bar{F}(G)^3\big] = \mathbb{E}\big[(F(G) - m)^3\big]\,.$$

This proves (3.1). If the third central moment vanished in this case, then (3.1) would give $\rho(G) = 0$ almost surely, hence $\rho \equiv 0$ by continuity and full support, contrary to the assumption. So the third central moment is strictly positive whenever F is not affine. □

3.2 Proof of Theorem A

Two results of [BCC+26] identify the law of $\mathscr{H}_t^{\vartheta}(\psi)$ with that of a convex function of a Gaussian variable, and a third excludes affine functions. A real random variable X is of *Ehrhard type* [BCC+26, Definition 3.5] if either X is constant, or $\mathbb{P}(X < y) < 1$ for every $y \in \mathbb{R}$ and $y \mapsto \Phi^{-1}(\mathbb{P}(X < y))$ is concave, where Φ is the standard Gaussian distribution function. By [BCC+26, Lemma 3.6], which collects results of [FMP25, LX26], a real random variable is of Ehrhard type if and only if

$$X \overset{\mathrm{d}}{=} F(G) \qquad \text{for some nondecreasing convex } F : \mathbb{R} \to \mathbb{R}\,. \tag{3.4}$$

By [BCC+26, Proposition 3.8], $\mathscr{H}_t^{\vartheta}(\psi)$ is of Ehrhard type for every $t > 0$, $\vartheta \in \mathbb{R}$ and nonnegative $\psi \in C_c^{\infty}(\mathbb{R}^2)$. Finally, Theorem 1.6 of [BCC+26] states that for nonnegative ψ that is not identically zero, $\mathscr{H}_t^{\vartheta}(\psi)$ is not Gaussian, degenerate Gaussian laws included.

Proof of Theorem A. Let $X = \mathscr{H}_t^{\vartheta}(\psi)$. By (2.2), $\mathbb{E}|X|^3 < \infty$. By [BCC+26, Proposition 3.8] and (3.4) there is a nondecreasing convex $F : \mathbb{R} \to \mathbb{R}$ with $X \overset{\mathrm{d}}{=} F(G)$. Then $\mathbb{E}|F(G)|^3 = \mathbb{E}|X|^3 < \infty$, and X and $F(G)$ have the same third central moment. If F were affine, $F(x) = \alpha x + \beta$ with $\alpha \geq 0$, the law of X would be $\mathcal{N}(\beta, \alpha^2)$, which [BCC+26, Theorem 1.6] excludes. Proposition 3.1 therefore gives $\mathbb{E}[(X - \mathbb{E}X)^3] > 0$. □

3.3 Remarks

Remark 3.4 (Distance to the Gaussian law). With F as in the proof above, the pair $(F(G) - \mathbb{E}X, \sigma G)$ with $\sigma^2 = \operatorname{Var} X$ is a coupling of the law of $X - \mathbb{E}X$ and $\mathcal{N}(0, \operatorname{Var} X)$. Proposition 3.1 therefore bounds the Wasserstein distance of order three:

$$W_3\big(\mathcal{L}(X - \mathbb{E}X), \mathcal{N}(0, \operatorname{Var} X)\big)^3 \leq 2\,\mathbb{E}\big[(X - \mathbb{E}X)^3\big]\,.$$

Thus a third central moment that is small compared with $(\operatorname{Var} X)^{3/2}$ forces X to be close to a Gaussian variable. Applied to the normalised averages in (2.4), this bound would give a rate of convergence from an upper bound on their third central moments, which we do not have. Nor do we have a lower bound on $\mathbb{E}[(X - \mathbb{E}X)^3]$ in terms of t, ϑ and ψ; the Ehrhard class itself contains laws with arbitrarily small positive skewness, such as $G + \eta \max(G, 0)$ for small $\eta > 0$.

Remark 3.5 (Higher odd moments). For X as in Theorem A, the same argument shows that $\mathbb{E}[(X - \mathbb{E}X)^{2k+1}] > 0$ for every integer $k \geq 1$; the weak inequality is again contained in [vZ64, Theorem 2.2.1]. Write $X \overset{\mathrm{d}}{=} F(G)$ as above. Since F is not affine, we are in the last case of the proof of Proposition 3.1, with $\sigma > 0$, $\bar{F}$, crossing points $x_1 < x_2$ and values $a_i = \sigma x_i$ as there; the interval (x_1, x_2) has positive Gaussian measure. Replace P by

$$P_k(y) = y^{2k+1} - \tfrac{1}{2} b_k y^2 - b_k' y\,,$$

where $y \mapsto b_k y + b_k'$ is the secant line through a_1 and a_2 of the strictly convex function $y \mapsto (2k+1)y^{2k}$. Then P_k' is negative on (a_1, a_2) and positive outside $[a_1, a_2]$, so $P_k(\bar{F}(x)) \geq P_k(\sigma x)$ for every x, with strict inequality on (x_1, x_2). Since X has moments of all orders by (2.2), we may take expectations; the quadratic and linear terms again cancel, and the odd moment of σG vanishes.

4 Response to the initial condition

Theorem B rests on one observation. At a positive smoothing parameter r, the function $q \mapsto \log \mathscr{Z}_{0,t}^{\vartheta}(\mathrm{e}^{h+qv}, g_r^{(x)})$ is the logarithmic moment generating function of v under a probability measure

on $\mathbb{R}^2$, the law of the starting point of a polymer path that ends near x at time t. Its derivatives are therefore cumulants of the bounded function v, and they are bounded by powers of $M = \|v\|_{L^\infty}$ uniformly in r, x and the realisation. Corollary 6.2 of [BCC+26] identifies the limit as $r \downarrow 0$ of $\mathcal{R}_r(q)$ for each fixed q, and the uniform bounds on three derivatives convert this convergence at fixed q into convergence of $\mathcal{R}_r$ and its first two derivatives, uniformly on compact sets of q.

Throughout this section $t > 0$ and $\vartheta \in \mathbb{R}$ are fixed, $h, v : \mathbb{R}^2 \to \mathbb{R}$ are bounded and continuous, $\psi \in C_c^\infty(\mathbb{R}^2)$, and $A = \|\psi\|_{L^1(\mathbb{R}^2)}$, $M = \|v\|_{L^\infty(\mathbb{R}^2)}$.

4.1 Positive smoothing

We first check that $\mathcal{R}_r$ is well defined. Fix $r > 0$. The heat kernel satisfies $g_r(y' - x) \le 2\mathrm{e}^{|x|^2/(2r)} g_{2r}(y')$, because $|y'|^2 \le 2|y' - x|^2 + 2|x|^2$. On the event Ω_r that $\mathscr{Z}^\vartheta_{0,t}(\mathbf{1}, g^{(0)}_{2r}) < \infty$ and that $\mathscr{Z}^\vartheta_{0,t}$ is not the zero measure, dominated convergence therefore shows that, for every bounded continuous h_0, the function $x \mapsto \mathscr{Z}^\vartheta_{0,t}(\mathrm{e}^{h_0}, g^{(x)}_r)$ is finite and continuous on $\mathbb{R}^2$; it is strictly positive everywhere, since $g_r > 0$. The event Ω_r has probability one: the first condition holds because $\mathbb{E}\mathscr{Z}^\vartheta_{0,t}(\mathbf{1}, g^{(0)}_{2r}) = 1$, and the second because $\mathscr{Z}^\vartheta_{0,t}(\mathbf{1}, g^{(0)}_r) > 0$ almost surely, by [BCC+26, Lemma 5.2] and the time reversal [BCC+26, (5.2)]. For each fixed x the function is measurable in the realisation. Hence $(\omega, x) \mapsto \log \mathscr{Z}^\vartheta_{0,t}(\mathrm{e}^{h_0}, g^{(x)}_r)$ is jointly measurable, and on Ω_r it is continuous, hence locally bounded, in x. The same lemma bounds the logarithmic moments, so that the centring $\mathbb{E}\log \mathscr{Z}^\vartheta_{0,t}(\mathbf{1}, g^{(0)}_r)$ is finite. On Ω_r the measures

$$\pi^x_{r,q}(\mathrm{d}y) = \frac{\mathrm{e}^{h(y)+qv(y)}\, \mathscr{Z}^\vartheta_{0,t}\big(\mathrm{d}y, g^{(x)}_r\big)}{\mathscr{Z}^\vartheta_{0,t}\big(\mathrm{e}^{h+qv}, g^{(x)}_r\big)}\,, \tag{4.1}$$

are therefore probability measures on $\mathbb{R}^2$ for every $x \in \mathbb{R}^2$ and $q \in \mathbb{R}$. We write $\pi^x_{r,q}(v)$ and $\mathrm{Var}_{\pi^x_{r,q}}(v)$ for the mean and the variance of v under $\pi^x_{r,q}$.

Proposition 4.1. *Let $r > 0$. Almost surely the function $\mathcal{R}_r$ of (1.2) is well defined and three times continuously differentiable on $\mathbb{R}$, and for every $q \in \mathbb{R}$*

$$\mathcal{R}_r'(q) = \int_{\mathbb{R}^2} \psi(x)\, \pi^x_{r,q}(v)\, \mathrm{d}x\,, \qquad \mathcal{R}_r''(q) = \int_{\mathbb{R}^2} \psi(x)\, \mathrm{Var}_{\pi^x_{r,q}}(v)\, \mathrm{d}x\,, \qquad |\mathcal{R}_r'''(q)| \le 8AM^3\,, \tag{4.2}$$

where both integrands are integrable. Moreover, almost surely, for every $q \in \mathbb{R}$,

$$|\mathcal{R}_r(q)| \le AM|q|\,, \qquad |\mathcal{R}_r'(q)| \le AM\,, \qquad |\mathcal{R}_r''(q)| \le AM^2\,, \tag{4.3}$$

and $\mathcal{R}_r''(q) \ge 0$ if $\psi \ge 0$.

Proof. Work on the event Ω_r and fix $x \in \mathbb{R}^2$. Dividing numerator and denominator by $\mathscr{Z}^\vartheta_{0,t}(\mathrm{e}^h, g^{(x)}_r)$ shows that the integrand of (1.2) is

$$\Lambda_x(q) = \log \mathscr{Z}^\vartheta_{0,t}\big(\mathrm{e}^{h+qv}, g^{(x)}_r\big) - \log \mathscr{Z}^\vartheta_{0,t}\big(\mathrm{e}^h, g^{(x)}_r\big) = \log \int_{\mathbb{R}^2} \mathrm{e}^{qv(y)}\, \pi^x_{r,0}(\mathrm{d}y)\,,$$

the logarithmic moment generating function of v under $\pi^x_{r,0}$. Since $|v| \le M$, differentiation under the integral is allowed to every order, and the n-th derivative of Λ_x at q is the n-th cumulant of v under the tilted measure $\mathrm{e}^{qv}\pi^x_{r,0}/\pi^x_{r,0}(\mathrm{e}^{qv}) = \pi^x_{r,q}$. The first three cumulants are the mean, the variance and the third central moment, and $|v| \le M$ gives

$$|\Lambda_x(q)| \le M|q|\,, \qquad |\Lambda_x'(q)| = |\pi^x_{r,q}(v)| \le M\,, \qquad 0 \le \Lambda_x''(q) \le M^2\,, \qquad |\Lambda_x'''(q)| \le (2M)^3\,,$$

the last because $|v - \pi^x_{r,q}(v)| \le 2M$. These bounds hold for all q and x. Since ψ is integrable, dominated convergence allows differentiating $\int \psi(x)\Lambda_x(q)\,\mathrm{d}x$ three times under the integral, which gives (4.2), and integrating the bounds against $|\psi|$ gives (4.3). If $\psi \ge 0$, then $\mathcal{R}''_r(q)$ is an integral of variances against a nonnegative weight and hence nonnegative. □

4.2 From fixed parameters to derivatives

The convergence supplied by [BCC+26] concerns $\mathcal{R}_r(q)$ at one value of q at a time. The next lemma uses the uniform bounds of Proposition 4.1 to compare derivatives through finite differences: a difference quotient over a step ε approximates the derivative up to an error of order ε, controlled by the next derivative, while dividing by ε costs only a factor ε^{-1} on the difference of the functions.

Lemma 4.2. *Let $a \le b$, $\varepsilon > 0$ and $B, E \ge 0$, and let f, g be real functions.*

(a) If $f, g \in C^2([a, b+\varepsilon])$, $|f - g| \le E$ and $|f''|, |g''| \le B$ on $[a, b+\varepsilon]$, then, on $[a, b]$, $|f' - g'| \le 2E/\varepsilon + B\varepsilon$.

(b) If $f, g \in C^3([a, b+2\varepsilon])$, $|f - g| \le E$ and $|f'''|, |g'''| \le B$ on $[a, b+2\varepsilon]$, then, on $[a, b]$, $|f'' - g''| \le 4E/\varepsilon^2 + 2B\varepsilon$.

Proof. For $x \in [a, b]$, Taylor's formula with integral remainder gives

$$f(x+\varepsilon) - f(x) - \varepsilon f'(x) = \int_0^\varepsilon (\varepsilon - u)\, f''(x+u)\,\mathrm{d}u\,,$$

so that $|f'(x) - \varepsilon^{-1}(f(x+\varepsilon) - f(x))| \le B\varepsilon/2$, and similarly for g. Since the difference quotient of $f - g$ is at most $2E/\varepsilon$ in absolute value, (a) follows. For (b), two applications of the fundamental theorem of calculus give

$$\frac{f(x+2\varepsilon) - 2f(x+\varepsilon) + f(x)}{\varepsilon^2} = \int_0^1\!\!\int_0^1 f''\big(x + \varepsilon(s+s')\big)\,\mathrm{d}s\,\mathrm{d}s'\,.$$

Since $|f''(x+\varepsilon(s+s')) - f''(x)| \le B\varepsilon(s+s')$ and $\int_0^1\int_0^1 (s+s')\,\mathrm{d}s\,\mathrm{d}s' = 1$, the second difference quotient differs from $f''(x)$ by at most $B\varepsilon$, and similarly for g. The second difference quotient of $f - g$ is at most $4E/\varepsilon^2$ in absolute value, which gives (b). □

4.3 Proof of Theorem B

We first quote from [BCC+26] that smoothing the position at time t, instead of removing the disorder of $[t-\delta, t]$, yields the same field. For bounded continuous h_0, $\varphi \in C_c^\infty(\mathbb{R}^2)$ and $r > 0$, put

$$\widetilde{\mathscr{H}}^{\vartheta,(r)}_t(h_0, \varphi) = \int_{\mathbb{R}^2} \varphi(x)\Big\{\log \mathscr{Z}^\vartheta_{0,t}\big(\mathrm{e}^{h_0}, g^{(x)}_r\big) - \mathbb{E}\log \mathscr{Z}^\vartheta_{0,t}\big(\mathbf{1}, g^{(0)}_r\big)\Big\}\,\mathrm{d}x\,.$$

Corollary 6.2 of [BCC+26] states that $\widetilde{\mathscr{H}}^{\vartheta,(r)}_t(h_0, \cdot) \to \mathscr{H}^\vartheta_t(h_0, \cdot)$ in probability in $H^{-a}_{\mathrm{loc}}(\mathbb{R}^2)$ for every $a > 0$, as $r \downarrow 0$. Testing against φ is continuous on this space, so

$$\widetilde{\mathscr{H}}^{\vartheta,(r)}_t(h_0, \varphi) \xrightarrow[r\downarrow 0]{\mathbb{P}} \mathscr{H}^\vartheta_t(h_0, \varphi)\,. \tag{4.4}$$

The centring in $\widetilde{\mathscr{H}}^{\vartheta,(r)}_t$ does not depend on h_0; this is what makes the response at positive r an exact difference of two such fields.

Proof of Theorem B. Step 1: convergence at a fixed parameter. Fix $q \in \mathbb{R}$. By definition, $\mathcal{R}_r(q) = \widetilde{\mathscr{H}}_t^{\vartheta,(r)}(h+qv,\psi) - \widetilde{\mathscr{H}}_t^{\vartheta,(r)}(h,\psi)$, because the two centrings cancel. Applying (4.4) to $h_0 = h+qv$ and to $h_0 = h$, both bounded and continuous, gives

$$\mathcal{R}_r(q) \xrightarrow[r\downarrow 0]{\mathbb{P}} \mathscr{H}_t^{\vartheta}(h+qv,\psi) - \mathscr{H}_t^{\vartheta}(h,\psi)\,. \tag{4.5}$$

Since $|\mathcal{R}_r(q)| \le AM|q|$ by (4.3), the limit obeys the same bound and the convergence holds in $L^p(\mathbb{P})$ for every $p \in [1,\infty)$. In particular, for every q and p, the family $(\mathcal{R}_r(q))_{r>0}$ is Cauchy in $L^p(\mathbb{P})$ as $r \downarrow 0$.

Step 2: uniform convergence on compact intervals. Fix $a \le b$ and $p \in [1,\infty)$, and write $\|Y\|_p = \mathbb{E}[|Y|^p]^{1/p}$. For $r, r' > 0$ and a finite set $\Sigma \subset [a,b]$ such that every point of $[a,b]$ lies within distance η of Σ, the bound $|\mathcal{R}_r'| \le AM$ gives

$$\Big\| \sup_{[a,b]} |\mathcal{R}_r - \mathcal{R}_{r'}| \Big\|_p \le \sum_{q\in\Sigma} \|\mathcal{R}_r(q) - \mathcal{R}_{r'}(q)\|_p + 2AM\eta\,.$$

Letting first $r, r' \downarrow 0$ and then $\eta \downarrow 0$ shows that the restrictions of $\mathcal{R}_r$ to $[a,b]$ are Cauchy in $L^p(\mathbb{P}; C([a,b]))$. This holds for every compact interval.

Step 3: derivatives. Fix $a \le b$, p and $\varepsilon \in (0,1]$. Lemma 4.2, applied pathwise to $f = \mathcal{R}_r$, $g = \mathcal{R}_{r'}$ with the bounds $B = AM^2$ in part (a) and $B = 8AM^3$ in part (b) from Proposition 4.1, gives

$$\begin{aligned}
\Big\| \sup_{[a,b]} |\mathcal{R}_r' - \mathcal{R}_{r'}'| \Big\|_p &\le \frac{2}{\varepsilon} \Big\| \sup_{[a,b+2]} |\mathcal{R}_r - \mathcal{R}_{r'}| \Big\|_p + AM^2\varepsilon\,,\\
\Big\| \sup_{[a,b]} |\mathcal{R}_r'' - \mathcal{R}_{r'}''| \Big\|_p &\le \frac{4}{\varepsilon^2} \Big\| \sup_{[a,b+2]} |\mathcal{R}_r - \mathcal{R}_{r'}| \Big\|_p + 16AM^3\varepsilon\,.
\end{aligned}$$

By Step 2 applied on $[a,b+2]$, the first terms tend to zero as $r, r' \downarrow 0$ for fixed ε; then let $\varepsilon \downarrow 0$. Hence the restrictions of $\mathcal{R}_r$ to $[a,b]$ are Cauchy in $L^p(\mathbb{P}; C^2([a,b]))$, where $C^2([a,b])$ carries the norm $\max_{j\le 2} \sup_{[a,b]} |f^{(j)}|$. These restrictions are random variables in this separable Banach space. Indeed, set $\mathcal{R}_r = 0$ off Ω_r; each $\mathcal{R}_r(q)$ is measurable by Fubini's theorem, $\mathcal{R}_r'(q)$ and $\mathcal{R}_r''(q)$ are limits of difference quotients, and the evaluations of f, f' and f'' at rational points of $[a,b]$ generate the Borel σ-algebra of $C^2([a,b])$. The same remark applies to $C^2(\mathbb{R})$ in Step 4.

Step 4: the limit. By completeness, for each $n \ge 1$ there is a random variable $\mathcal{R}^{[n]}$ in $C^2([-n,n])$ with $\mathcal{R}_r \to \mathcal{R}^{[n]}$ in $L^p(\mathbb{P}; C^2([-n,n]))$ for every $p \in [1,\infty)$; the limit does not depend on p, since convergence in L^p implies convergence in probability. Restricting to $[-n,n]$ the limit on $[-n-1, n+1]$ gives $\mathcal{R}^{[n]}$, almost surely. Hence there is a random variable $\mathcal{R}$ in $C^2(\mathbb{R})$ whose restriction to each $[-n,n]$ equals $\mathcal{R}^{[n]}$ almost surely, and every compact interval lies in some $[-n,n]$, which proves (iii). Along a sequence $r_k \downarrow 0$ chosen by a diagonal argument, $\mathcal{R}_{r_k} \to \mathcal{R}$ in $C^2([-n,n])$ for every n, almost surely.

Step 5: properties of the limit. For fixed q, $\mathcal{R}_r(q) \to \mathcal{R}(q)$ in $L^1(\mathbb{P})$ by Step 4, and (4.5) identifies the limit, which proves (i). Along the sequence r_k the bounds (4.3), and the sign of $\mathcal{R}_r''$ when $\psi \ge 0$, hold simultaneously for all q and pass to the limit, which proves (ii).

Step 6: uniqueness. Let $\widetilde{\mathcal{R}}$ be another random variable in $C^2(\mathbb{R})$ satisfying (i). Then $\widetilde{\mathcal{R}}(q) = \mathcal{R}(q)$ almost surely for each rational q, hence almost surely for all rational q simultaneously, hence for all q by continuity. Two C^2 functions that agree everywhere have the same derivatives, so $\widetilde{\mathcal{R}} = \mathcal{R}$ in $C^2(\mathbb{R})$ almost surely. □

Along the sequence r_k of Step 4, the bound $|\mathcal{R}_r'''| \le 8AM^3$ passes to the limit as a Lipschitz bound: almost surely $|\mathcal{R}''(q) - \mathcal{R}''(q')| \le 8AM^3|q - q'|$ for all q, q'. This gives the following description of the response as a map into L^p.

Corollary 4.3. *Let $p \in [1, \infty)$. The map $q \mapsto \mathscr{H}_t^\vartheta(h + qv, \psi) - \mathscr{H}_t^\vartheta(h, \psi)$ from $\mathbb{R}$ to $L^p(\mathbb{P})$ is twice differentiable, and its first and second derivatives at q are $\mathcal{R}'(q)$ and $\mathcal{R}''(q)$.*

Proof. By Theorem B(i), the map takes the value $\mathcal{R}(q)$ at q. Taylor's formula and the bounds of Theorem B(ii) give, almost surely, $|\mathcal{R}(q + u) - \mathcal{R}(q) - u\mathcal{R}'(q)| \le \frac{1}{2}AM^2u^2$ for all u, and the Lipschitz bound above gives $|\mathcal{R}'(q + u) - \mathcal{R}'(q) - u\mathcal{R}''(q)| \le 4AM^3u^2$. Dividing by u and taking L^p norms, both difference quotients converge in $L^p(\mathbb{P})$ as $u \to 0$. □

4.4 Remarks

Remark 4.4 (Higher derivatives). The n-th cumulant of a random variable bounded by M is at most a_nM^n in absolute value, where $a_n = \sum_{\mathcal{P}}(|\mathcal{P}| - 1)!$ is the sum over all partitions $\mathcal{P}$ of $\{1, \dots, n\}$. Since $(|\mathcal{P}| - 1)! \le |\mathcal{P}|!$, a_n is at most the number of ordered partitions of $\{1, \dots, n\}$, which inject into the maps from $\{1, \dots, n\}$ to itself by sending i to the index of its block; hence $a_n \le n^n \le n!\,\mathrm{e}^n$. The proof of Proposition 4.1 therefore gives $|\mathcal{R}_r^{(n)}| \le Aa_nM^n$ for every $n \ge 1$, and iterated finite differences extend Lemma 4.2 to every order. Running Steps 3–5 with these bounds shows that $\mathcal{R}$ is infinitely differentiable, that $\mathcal{R}_r \to \mathcal{R}$ in $L^p(\mathbb{P}; C^k([a, b]))$ for every k, and, through the Taylor remainder bound $A(\mathrm{e}M|u|)^{N+1}$, that $\mathcal{R}$ is real analytic. We do not use this.

Remark 4.5 (Starting-point laws). Theorem B and (4.2) express $\mathcal{R}'(q)$ and $\mathcal{R}''(q)$ as limits in $L^p(\mathbb{P})$ of $\int \psi(x)\,\pi_{r,q}^x(v)\,\mathrm{d}x$ and of $\int \psi(x)\,\mathrm{Var}_{\pi_{r,q}^x}(v)\,\mathrm{d}x$. They say nothing about convergence of the measures $\pi_{r,q}^x$ themselves. If, for almost every x, these laws converged weakly to a limit π_q^x, then $\mathcal{R}'(q)$ and $\mathcal{R}''(q)$ would be the ψ-weighted integrals of the mean and the variance of v under π_q^x; we do not know whether this holds. Without it, a variance description of $\mathcal{R}''(q)$ can fail for averaged laws. Suppose that $\psi \ge 0$ and that there were a sequence $r_k \downarrow 0$ with $\pi_{r_{2k-1},q}^x = \pi_1^x$ and $\pi_{r_{2k},q}^x = \pi_2^x$ for every x, where $\pi_1^x(v) \ne \pi_2^x(v)$ on a set of positive Lebesgue measure on which $\psi > 0$. Then Theorem B(iii) forces $\mathcal{R}''(q) = \int \psi(x)\,\mathrm{Var}_{\pi_1^x}(v)\,\mathrm{d}x = \int \psi(x)\,\mathrm{Var}_{\pi_2^x}(v)\,\mathrm{d}x$, whereas the averages $(2N)^{-1}\sum_{k \le 2N} \pi_{r_k,q}^x$ equal $\frac{1}{2}(\pi_1^x + \pi_2^x)$, under which the corresponding integral of variances is strictly larger, by the law of total variance.

5 Brownian motion across scales

Theorem C has two parts: convergence of the finite-dimensional distributions of X_L, and tightness of its laws in $C([0, S])$. They rest on different structures. This section gives the argument in outline. Proposition 5.1, Lemma 5.4, the deductions from Lemma 5.3, the joint Gaussian limit, the refinement of windows and the final assembly in Section 5.4 are proved completely. For Lemma 5.3, the smooth band fields, Lemma 5.6, the maximal inequality and the remainder bounds (5.13) we indicate the argument and the estimates of [BCC+26] it uses.

Independent time blocks. After a time reversal, $\mathscr{H}_1^\vartheta(\varphi_\varepsilon^{(0)})$ is close to the logarithm of the stochastic heat flow started from a heat kernel of variance ε^2 at time ε^2 and run up to time one. Cut the time interval $[\varepsilon^2, 1]$ at the geometric sequence ρ^{-j}, for a large fixed ratio ρ. Over each interval $[\rho^{-j}, \rho^{-j+1}]$ a polymer started from a heat profile at scale $\rho^{-j/2}$ spreads to scale $\rho^{-(j-1)/2}$, the scale of the next profile, so the partition function nearly factorises into a product of independent

masses Z_j, one for each interval. By the scaling covariance (2.3), the j-th interval becomes the fixed interval $[1,\rho]$ at parameter $\vartheta - j\log\rho$, far below the critical window, and $\log Z_j$ has variance of order $1/j$; partial sums of n blocks therefore have variance of order $\log n$. Averages at the scales $\varepsilon_L(s)$, $s\in[0,S]$, correspond to nested partial sums of the single independent sequence $(\log Z_j)_j$, suitably truncated, and the number of terms grows like e^{Ls}. Nested partial sums of one sequence of independent bounded variables have jointly Gaussian limits with independent increments, which is Brownian motion; Gaussian marginals and the right covariance alone would not give this. The variance of the partial sums is identified through the scalar central limit theorem (5.5) below.

Spatial stationarity. The approximation by partial sums holds at each fixed scale with an error that is small only in probability, and our estimates do not make it uniform over the continuum of scales in a window. Tightness comes instead from spatial stationarity. Within a window of scales, the averages $\mathscr{H}(\varphi_r^{(0)})$ are, up to a term independent of the width r and controlled remainders, spatial averages $\int\varphi(z)f(rz)\,\mathrm{d}z$ of one stationary field f that keeps only the frequencies between the two ends of the window. A stationary version of the Hardy–Littlewood maximal inequality bounds all these averages at once by the size of f at one point, and a truncated form of it keeps the Gaussian tail that is needed when the windows are refined.

Throughout this section $t>0$, $\vartheta\in\mathbb{R}$, $x\in\mathbb{R}^2$ and $S>0$ are fixed, $\varphi\in C_c^\infty(\mathbb{R}^2)$ is nonnegative with $\int\varphi=1$, $\varphi_r=\varphi_r^{(0)}$, and $\varepsilon_L(s)=\exp(-\mathrm{e}^{Ls})$. We write $\mathscr{H}=\mathscr{H}_t^\vartheta$ for the flat field, r for widths and s,s' for scale times, and note that

$$\log\log\frac{1}{\varepsilon_L(s)}=Ls\,. \tag{5.1}$$

5.1 Covariance on the doubly logarithmic scale

The first ingredient is a consequence of the covariance asymptotic (2.5) that requires no further information about the field. It identifies the covariance of X_L with that of Brownian motion and allows us to change the width and the shape of the mollifier at a cost that is negligible on the scale $\sqrt{L}$.

Proposition 5.1. *Let $t>0$ and $\vartheta\in\mathbb{R}$. Let $\varphi,\psi\in C_c^\infty(\mathbb{R}^2)$ be nonnegative with $\int\varphi=\int\psi=1$, and let $a_L,b_L>0$ satisfy $L^{-1}\log\log(1/a_L)\to s$ and $L^{-1}\log\log(1/b_L)\to s'$ as $L\to\infty$, for some $s,s'>0$. Then*

$$\lim_{L\to\infty}\frac{1}{L}\operatorname{Cov}\bigl(\mathscr{H}_t^\vartheta(\varphi_{a_L}^{(x)}),\mathscr{H}_t^\vartheta(\psi_{b_L}^{(x)})\bigr)=\min(s,s')\,.$$

For $a_L=\varepsilon_L(s)$ and $b_L=\varepsilon_L(s')$, the convergence is uniform in $s,s'\in[s_0,S]$ for every $s_0>0$.

Proof. Let $a\ge b>0$ and $\varrho=b/a\in(0,1]$. By (2.5) and a change of variables,

$$\operatorname{Cov}\bigl(\mathscr{H}(\varphi_a^{(x)}),\mathscr{H}(\psi_b^{(x)})\bigr)=\mathbb{E}\bigl[\mathsf{C}_t^\vartheta\bigl(a(U-\varrho V)\bigr)\bigr]\,,$$

where U and V are independent random vectors with densities φ and ψ. For every $\varrho\in[0,1]$, $U-\varrho V$ has the density $w\mapsto\int\varphi(w+\varrho v)\psi(v)\,\mathrm{d}v$, which is bounded by $\|\varphi\|_\infty$ and supported in a fixed ball $\{|w|\le K\}$. Hence $\sup_{0\le\varrho\le1}\mathbb{E}\log^+(1/|U-\varrho V|)<\infty$: the singularity of the kernel at the moving diagonal $\{U=\varrho V\}$ is integrable uniformly in ϱ. The elementary inequalities

$$\log\Bigl(\log\frac{1}{a}-\log^+K\Bigr)\le\log\log\frac{1}{a|w|}\le\log\log\frac{1}{a}+\frac{\log^+(1/|w|)}{\log(1/a)}\qquad(0<|w|\le K)$$

then give $\mathbb{E}\log\log(1/(a|U-\varrho V|)) = \log\log(1/a) + O(1/\log(1/a))$ uniformly in ϱ. Since every value of $a(U-\varrho V)$ lies in the ball of radius Ka, the diagonal asymptotic (2.5) yields

$$\operatorname{Cov}\big(\mathscr{H}(\varphi_a^{(x)}), \mathscr{H}(\psi_b^{(x)})\big) = \log\log(1/a)\,(1+o(1)) \qquad (a \to 0)\,,$$

uniformly in $b \le a$. For each L, apply this with the larger of a_L, b_L in the role of a, exchanging (φ, a_L) and (ψ, b_L) if necessary; the covariance is $\log\log(1/\max(a_L,b_L))(1+o(1))$, and $L^{-1}\log\log(1/\max(a_L,b_L)) \to \min(s,s')$. For $a_L = \varepsilon_L(s)$, the error is controlled by the behaviour of C_t^ϑ on the ball of radius $K\varepsilon_L(s_0)$, which gives the uniformity. □

Proposition 5.1 gives the increments of X_L the covariance of Brownian motion and permits changes of width and mollifier. All averages of $\mathscr{H}$ are centred, since $\mathscr{H}$ is the limit in $L^1(\mathbb{P})$ of the centred fields (2.1). Applied to two variances and a cross covariance, the proposition therefore gives

$$\mathbb{E}\big[(X_L(s) - X_L(s'))^2\big] \to |s-s'| \qquad (s,s'>0)\,, \tag{5.2}$$

$$\frac{1}{L}\mathbb{E}\Big[\big(\mathscr{H}(\varphi_{a_L}^{(x)}) - \mathscr{H}(\psi_{b_L}^{(x)})\big)^2\Big] \to 0 \qquad \text{if } s = s'\,. \tag{5.3}$$

The second statement allows replacing the width $\varepsilon_L(s)$ by any width a_L with $\log\log(1/a_L) = Ls + o(L)$, and the mollifier φ by any other ψ, jointly for finitely many s.

5.2 Finite-dimensional distributions

Proposition 5.2. *For every finite set $I \subset [0,\infty)$, $(X_L(s))_{s\in I} \Rightarrow (B_s)_{s\in I}$ as $L \to \infty$, where B is a standard Brownian motion.*

We prove Proposition 5.2 in five steps: a reduction, a time reversal, the block approximation, the identification of the centring, and the joint limit.

Reduction. We reduce to $t=1$, $x=0$, a centred mollifier and geometric widths. If $0 \in I$, then $X_L(0) = L^{-1/2}\mathscr{H}(\varphi_{\mathrm{e}^{-1}}^{(x)}) \to 0$ in $L^2(\mathbb{P})$, and Slutsky's theorem reduces the claim to $I \subset (0,\infty)$. By the scaling covariance (2.3) with $\kappa = t$, applied jointly to finitely many tests,

$$\big(\mathscr{H}_t^\vartheta(\varphi_{\varepsilon_i}^{(x)})\big)_i \overset{\mathrm{d}}{=} \big(\mathscr{H}_1^{\vartheta+\log t}(\varphi_{\varepsilon_i/\sqrt{t}}^{(x/\sqrt{t})})\big)_i\,,$$

and $\log\log(\sqrt{t}/\varepsilon_L(s)) = Ls + o(L)$. Proposition 5.1 holds for every t and ϑ, so (5.3) applies to $\mathscr{H}_1^{\vartheta+\log t}$: the widths may be changed to any widths with the same doubly logarithmic size, and φ may be replaced by a nonnegative mollifier of integral one with barycentre zero. By translation invariance we may take $x=0$. Writing ϑ again for $\vartheta + \log t$, it suffices to treat $t=1$, a centred mollifier, and the widths $\rho^{-n_L(s)/2}$, where $\rho > 1$ is a large fixed ratio and

$$n_L(s) = \Big\lfloor \frac{2\mathrm{e}^{Ls}}{\log\rho} \Big\rfloor\,, \qquad \log\log\rho^{n_L(s)/2} = \log n_L(s) + \log\tfrac{\log\rho}{2} = Ls + o(1)\,. \tag{5.4}$$

Time reversal. At $t=1$ we use the time-reversed fields of [BCC+26, Sections 2.2 and 8]. Put

$$F_\delta(y) = \log \mathscr{Z}_{\delta,1}^\vartheta\big(g_\delta^{(y)}, \mathbf{1}\big)\,, \qquad 0 < \delta < 1\,,$$

the logarithm of the flow started at time δ from a heat kernel of variance δ and run to time one. On the probability space of the flow there is a random distribution $\mathscr{H}$ with the law of

$\mathscr{H}_1^\vartheta$, the limit in $L^2(\mathbb{P})$ of $\int \varphi(y)\{F_\delta(y) - \mathbb{E}F_\delta(y)\}\,\mathrm{d}y$ as $\delta \downarrow 0$ [BCC+26, (2.8), (2.11)]. Two results of [BCC+26, Section 8] connect it with the variables $F_\delta(0)$: the scalar central limit theorem [BCC+26, Theorem 8.1], which relies on Gu and Tsai [GT26],

$$\frac{F_{\varepsilon^2}(0) - \mathbb{E}F_{\varepsilon^2}(0)}{\sqrt{\log\log(1/\varepsilon)}} \implies \mathcal{N}(0,1) \qquad (\varepsilon \downarrow 0)\,, \tag{5.5}$$

and the comparison [BCC+26, (8.6)], valid for nonnegative φ of integral one with barycentre zero: for some $\varepsilon_0 \in (0,1)$,

$$\sup_{0<\varepsilon<\varepsilon_0} \big\| \check{\mathscr{H}}(\varphi_\varepsilon) - \big(F_{\varepsilon^2}(0) - \mathbb{E}F_{\varepsilon^2}(0)\big) \big\|_{L^1(\mathbb{P})} < \infty\,. \tag{5.6}$$

Since $\check{\mathscr{H}}$ is one random distribution, (5.6) applies to finitely many widths simultaneously. With $\tau_n = \rho^{-n}$, the width $\rho^{-n/2}$ corresponds to $\delta = \varepsilon^2 = \tau_n$, and it therefore suffices to prove that $L^{-1/2}(F_{\tau_{n_L(s)}}(0) - \mathbb{E}F_{\tau_{n_L(s)}}(0))_{s\in I} \Rightarrow (B_s)_{s\in I}$.

The block approximation. Define the block masses

$$Z_j = \mathscr{Z}^\vartheta_{\tau_j,\tau_{j-1}}(g_{\tau_j}, \mathbf{1})\,, \qquad j \geq 1\,.$$

They are defined on the same flow and do not depend on n, and $\mathbb{E}Z_j = 1$. They are independent: the lattice partition functions over disjoint time blocks contain disjoint disorder and are independent [BCC+26, after (7.3)], and independence passes to the joint limit [BCC+26, Theorem A with (4.6)–(4.10)]. Put $Y_j = \mathbf{1}_{\{|Z_j-1|\leq 1/2\}} \log Z_j$, so that $|Y_j| \leq \log 2$, and $S_n = \sum_{j\leq n} Y_j$. For $n > \ell$, let

$$A_{n,\ell} = \mathscr{Z}^\vartheta_{\tau_n,\tau_\ell}(g_{\tau_n}, \mathbf{1})\,, \qquad R_{n,\ell} = \frac{A_{n,\ell}}{\prod_{j=\ell+1}^n Z_j}\,, \qquad Q_{n,\ell} = \frac{\exp F_{\tau_n}(0)}{A_{n,\ell}}\,,$$

so that $F_{\tau_n}(0) = \log Q_{n,\ell} + \log R_{n,\ell} + \sum_{j=\ell+1}^n \log Z_j$: the variable $Q_{n,\ell}$ accounts for the last time interval $[\tau_\ell, 1]$, and $R_{n,\ell}$ for the failure of the product formula before it. The heart of the argument is the following estimate.

Lemma 5.3. *If ρ is large enough, there are $C < \infty$ and $\ell_* \geq 1$ such that for all integers $n > \ell \geq \ell_*$, all $j \geq 1$ and all $u \geq 0$,*

$$\mathbb{E}(Z_j - 1)^2 \leq C/j\,, \qquad \mathbb{E}(Z_j - 1)^4 \leq C/j^2\,, \qquad \mathbb{P}\big(|R_{n,\ell} - 1| > \tfrac{1}{2}\big) \leq C/\ell\,,$$
$$\mathbb{P}(\log Q_{n,\ell} < -u) \leq C/\ell + C\ell \mathrm{e}^{-u}\,, \qquad \mathbb{P}(\log Q_{n,\ell} > u) \leq \mathrm{e}^{-u}\,.$$

Sketch of proof. Each estimate is proved for the lattice polymer partition functions of [BCC+26] at fixed n and ℓ, with constants independent of n, ℓ and the lattice size, and then passed to the flow: the finitely many masses involved converge jointly [BCC+26, Theorem A with (4.6)–(4.10)], their limits are almost surely positive [BCC+26, (5.10)], and the bounds pass to the limit by Fatou's lemma and the portmanteau theorem. Only after this transfer are n and ℓ allowed to grow. The moment bounds are [BCC+26, Lemma B.2], and the ratio bound follows from [BCC+26, Proposition 7.3].

The bound on $Q_{n,\ell}$ runs the argument of [BCC+26, proof of Lemma 7.8] from an arbitrary starting index, using [BCC+26, Lemma 7.7], which is stated for arbitrary consecutive blocks; the new point is that the constants do not depend on n. Let $\mathcal{F}_\ell$ be the σ-field of the disorder in the time interval $[\tau_n, \tau_\ell]$, and μ the polymer endpoint profile at time τ_ℓ started from the heat kernel at time

τ_n, normalised by $A_{n,\ell}$; then $Q_{n,\ell}$ is the mass over $[\tau_\ell, 1]$ started from μ. The expansion [BCC+26, (7.48)] writes μ as the heat profile at time τ_ℓ plus $R_{n,\ell}^{-1}$ times a signed correction of total mass zero. The heat profile contributes a mass with second moment $O(\ell)$, since the interval $[\tau_\ell, 1]$ consists of ℓ blocks [BCC+26, (4.4)]; this is the source of the factor ℓ. The correction is a sum over the blocks between τ_n and τ_ℓ of a factor carrying the earlier blocks, bounded in L^2 on the block event of [BCC+26, Proposition 7.3], times the open-string correction of the later blocks (the part of the expansion that has not closed by time τ_ℓ), bounded by [BCC+26, Proposition 7.4]; these two factors use disjoint blocks, and the resulting series is geometric once ρ is large, which gives $O(1/\ell)$ for the expected square of the seminorm that measures the effect of the correction on the future mass, uniformly in n, as in the proof of [BCC+26, (7.59)]. On the event $\mathsf{G} = \Omega^{\mathrm{b}} \cap \{|R_{n,\ell} - 1| \le \frac{1}{2}\}$ of [BCC+26, (7.51)], where Ω^{b} is the block event, the factor $R_{n,\ell}^{-1}$ is at most 2; $\mathsf{G} \in \mathcal{F}_\ell$, and $\mathbb{P}(\mathsf{G}^c) \le C/\ell$ by [BCC+26, (7.52)]. Hence

$$\mathbb{E}\big[\mathbf{1}_{\mathsf{G}}\, Q_{n,\ell}^2\big] \le C\ell\,.$$

At the level of the lattice partition functions, before the transfer, μ is deterministic given $\mathcal{F}_\ell$, the remaining disorder is independent Gaussian, $\log Q_{n,\ell}$ is a convex function of it, and $\mathbb{E}[Q_{n,\ell} \mid \mathcal{F}_\ell] = 1$. The Gaussian convexity inequality [BCC+26, (3.5)] with $p = 1$ gives $\mathbb{E}[Q_{n,\ell}^{-1} \mid \mathcal{F}_\ell] \le \mathbb{E}[Q_{n,\ell}^2 \mid \mathcal{F}_\ell]$. Multiplying by $\mathbf{1}_{\mathsf{G}}$ after conditioning, $\mathbb{E}[\mathbf{1}_{\mathsf{G}} Q_{n,\ell}^{-1}] \le C\ell$, and Markov's inequality gives the lower tail. The upper tail follows from $\mathbb{E}Q_{n,\ell} = 1$. □

On the event that $|R_{n,\ell} - 1| \le \frac{1}{2}$, $|\log Q_{n,\ell}| \le u$ and $|Z_j - 1| \le \frac{1}{2}$ for all $j > \ell$, we have $F_{\tau_n}(0) - S_n = \log Q_{n,\ell} + \log R_{n,\ell} - S_\ell$. The fourth moment bound controls the probability that some $|Z_j - 1| > \frac{1}{2}$ with $j > \ell$ by C/ℓ, and Lemma 5.3 gives, uniformly in $n > \ell \ge \ell_*$ and $u \ge 0$,

$$\mathbb{P}\big(|F_{\tau_n}(0) - S_n| > \ell \log 2 + \log 2 + u\big) \le C/\ell + (1 + C\ell)\mathrm{e}^{-u}\,. \tag{5.7}$$

With $\ell_L = \lfloor L^{1/4} \rfloor$ and $u_L = 3 \log \ell_L$, a union bound over the finitely many indices $n_L(s)$, $s \in I$, gives

$$\max_{s \in I} \frac{|F_{\tau_{n_L(s)}}(0) - S_{n_L(s)}|}{\sqrt{L}} \;\xrightarrow{\;\mathbb{P}\;}\; 0\,. \tag{5.8}$$

Since $\mathbb{E}(Z_j - 1)^2 \le C/j$, also $\operatorname{Var} S_n \le C(1 + \log n)$.

The centring. Convergence in probability in (5.8) does not allow us to subtract expectations: a rare large error may have a large mean. The scalar limit (5.5) identifies both the centring and the variance.

Lemma 5.4. *Let Y_j be independent random variables with $|Y_j| \le K_0$, $S_n = \sum_{j \le n} Y_j$, and $a_n \to \infty$ with $\operatorname{Var} S_n \le C a_n^2$. Let F_n be integrable random variables with $(F_n - S_n)/a_n \to 0$ in probability and $(F_n - \mathbb{E}F_n)/a_n \Rightarrow \mathcal{N}(0,1)$. Then $\operatorname{Var} S_n / a_n^2 \to 1$ and $(\mathbb{E}F_n - \mathbb{E}S_n)/a_n \to 0$.*

Proof. Put $\widetilde{S}_n = (S_n - \mathbb{E}S_n)/a_n$, $v_n = \operatorname{Var} \widetilde{S}_n$ and $d_n = (\mathbb{E}F_n - \mathbb{E}S_n)/a_n$, so that $(F_n - \mathbb{E}F_n)/a_n = \widetilde{S}_n - d_n + o_{\mathbb{P}}(1)$. Both $\widetilde{S}_n$ and the left side are tight, so (d_n) is bounded. Along any subsequence choose a further subsequence with $v_n \to v$ and $d_n \to d$. The centred summands $(Y_j - \mathbb{E}Y_j)/a_n$ are bounded by $2K_0/a_n \to 0$, so the Lindeberg–Feller theorem (or Chebyshev's inequality if $v = 0$) gives $\widetilde{S}_n \Rightarrow \mathcal{N}(0, v)$ along it. Then $(F_n - \mathbb{E}F_n)/a_n \Rightarrow \mathcal{N}(-d, v)$, which forces $d = 0$ and $v = 1$. □

We apply Lemma 5.4 with $F_n = F_{\tau_n}(0)$ and $a_n = \sqrt{\log n}$. The hypothesis on F_n is (5.5) at $\varepsilon = \rho^{-n/2}$, for which $\log\log(1/\varepsilon) = \log n + O(1)$, and (5.7) with $\ell = \lfloor (\log n)^{1/4} \rfloor$ and $u = 3 \log \ell$ gives $(F_n - S_n)/a_n \to 0$ in probability. Hence $\operatorname{Var} S_n \sim \log n$ and $F_{\tau_n}(0) - \mathbb{E}F_{\tau_n}(0) = S_n - \mathbb{E}S_n + o_{\mathbb{P}}(\sqrt{\log n})$.

The joint limit. For $s \le s'$, independence of the blocks gives $\mathrm{Cov}(S_{n_L(s)}, S_{n_L(s')}) = \mathrm{Var}\, S_{n_L(s)} \sim \log n_L(s) \sim Ls$. For real coefficients c_s, $s \in I$,

$$\frac{1}{\sqrt{L}} \sum_{s\in I} c_s \big(S_{n_L(s)} - \mathbb{E}S_{n_L(s)}\big) = \sum_j \frac{w_{j,L}}{\sqrt{L}} \big(Y_j - \mathbb{E}Y_j\big)\,, \qquad w_{j,L} = \sum_{s\in I:\, n_L(s)\ge j} c_s\,,$$

is a sum of independent summands bounded by $2\log 2 \sum_s |c_s|/\sqrt{L}$, with variance tending to $\sum_{s,s'} c_s c_{s'} \min(s,s')$. The Lindeberg–Feller theorem and the Cramér–Wold device give $L^{-1/2}(S_{n_L(s)} - \mathbb{E}S_{n_L(s)})_{s\in I} \Rightarrow (B_s)_{s\in I}$, and the previous two paragraphs transfer this to $F_{\tau_{n_L(s)}}(0) - \mathbb{E}F_{\tau_{n_L(s)}}(0)$. This proves Proposition 5.2.

5.3 Tightness

Tightness uses only spatial stationarity, the spectral asymptotic of the covariance, and the convergence of the finite-dimensional distributions and second moments. We state it for a general random distribution, with $\widehat{f}(\xi) = \int \mathrm{e}^{-i\xi\cdot y} f(y)\,\mathrm{d}y$.

Proposition 5.5. *Let D be a random distribution on $\mathbb{R}^2$ such that $\mathbb{E}[D(f)] = 0$ and $\mathbb{E}[D(f)^2] < \infty$ for every $f \in C_c^\infty(\mathbb{R}^2)$, whose finite families $(D(f_i))_i$ have laws invariant under translation of the f_i, and suppose that there is a locally finite measure ν on $\mathbb{R}^2$ with*

$$\mathbb{E}[D(f)D(f')] = \int \widehat{f}(\xi)\,\overline{\widehat{f'}(\xi)}\,\nu(\mathrm{d}\xi) \quad (f, f' \in C_c^\infty(\mathbb{R}^2))\,, \qquad \lim_{R\to\infty} \frac{\nu(\bar{B}(0,R))}{\log\log R} = 1\,. \tag{5.9}$$

Let $\varphi \in C_c^\infty(\mathbb{R}^2)$ be nonnegative with $\int\varphi = 1$, and put $X_L^D(s) = L^{-1/2} D(\varphi_{\varepsilon_L(s)})$. Assume that for every finite $I \subset (0,\infty)$, $(X_L^D(s))_{s\in I} \Rightarrow (B_s)_{s\in I}$; that $\mathbb{E}X_L^D(s)^2 \to s$ for $s > 0$; and that $\mathbb{E}(X_L^D(s') - X_L^D(s))^2 \to s' - s$ for $0 < s < s'$. Then, for every $S > 0$ and $\eta > 0$,

$$\lim_{h\downarrow 0} \limsup_{L\to\infty} \mathbb{P}\Big(\sup_{s,s'\in[0,S],\,|s-s'|\le h} |X_L^D(s) - X_L^D(s')| > \eta \Big) = 0\,.$$

The paths $s \mapsto X_L^D(s)$ are continuous, because $s \mapsto \varphi_{\varepsilon_L(s)}$ is continuous into $C_c^\infty(\mathbb{R}^2)$ with supports in a fixed compact set; this applies in particular to X_L in Theorem C.

We sketch the proof of Proposition 5.5 in four steps. Constants C, c may depend on φ and ν but not on L, on the windows, or on the parameter R introduced below.

Smooth band fields. We cut off the high frequencies of D smoothly. Put $N(R) = \nu(\bar{B}(0,R))$ and $V(a) = 1 + \log\log(\mathrm{e}^{\mathrm{e}} + a^{-1})$; the asymptotic in (5.9) gives $N(2^{k+1}/a) \le C(k+1)V(a)$ for all $k \ge 0$ and $0 < a \le 1$. Let $\chi \in C_c^\infty(\mathbb{R}^2)$ be radial with $0 \le \chi \le 1$, $\chi = 1$ on $\bar{B}(0,1)$ and $\chi = 0$ outside $B(0,2)$, and let ζ be the Schwartz function with $\widehat{\zeta} = \chi$. The isometry (5.9) extends D to Schwartz tests in $L^2(\mathbb{P})$, preserving stationarity, and the field $D_a(y) = D(\zeta_a^{(y)})$ has a smooth, jointly measurable, stationary version: its derivatives of order k have variance at most $(2/a)^{2k} N(2/a)$. The same isometry gives the following comparison of shapes at one scale.

Lemma 5.6. *Under the hypotheses of Proposition 5.5, for a Schwartz function m with $\mathsf{m}(0) = 0$, $\int |\mathsf{m}(\varepsilon\xi)|^2\,\nu(\mathrm{d}\xi) = o(\log\log(1/\varepsilon))$ as $\varepsilon \downarrow 0$. In particular, for fixed $\kappa' > 0$ and $s > 0$, $L^{-1/2}(D_{\kappa'\varepsilon_L(s)}(0) - D(\varphi_{\varepsilon_L(s)})) \to 0$ in $L^2(\mathbb{P})$.*

Sketch of proof. For $K' > 1$, split the integral at $|\varepsilon\xi| = 1/K'$ and $|\varepsilon\xi| = K'$. The inner part is at most $\sup_{|z|\le 1/K'} |\mathsf{m}(z)|^2 N(1/(K'\varepsilon))$, the middle part at most $\|\mathsf{m}\|_\infty^2 (N(K'/\varepsilon) - N(1/(K'\varepsilon))) = o(\log\log(1/\varepsilon))$, and the outer part, summed over dyadic shells with the bound on $N(2^{k+1}K'/\varepsilon)$ and the decay of m, is at most $C_k K'^{-k} \log\log(1/\varepsilon)$ for every k. Let $\varepsilon \downarrow 0$ and then $K' \to \infty$. The second statement takes $\mathsf{m}(z) = \chi(\kappa' z) - \widehat{\varphi}(z)$ and uses (5.1). □

A stationary maximal inequality. Let $\mathcal{M}g(y) = \sup_{r>0} |B(y,r)|^{-1} \int_{B(y,r)} |g|$ be the Hardy–Littlewood maximal function, and $\mathcal{M}_{\le \bar r}$ the same supremum over $r \in (0, \bar r]$. For a stationary jointly measurable field g with $\mathbb{E}|g(0)| < \infty$,

$$\mathbb{P}\big(\mathcal{M}g(0) > \Lambda\big) \le C\Lambda^{-1}\mathbb{E}|g(0)| \qquad (\Lambda > 0)\,. \tag{5.10}$$

To see this, apply the deterministic weak $(1,1)$ inequality to $g\mathbf{1}_{[-k-\bar r, k+\bar r]^2}$ and take expectations; by stationarity, the expected measure of the set of $y \in [-k,k]^2$ where $\mathcal{M}_{\le \bar r} g(y) > \Lambda$ equals $(2k)^2 \mathbb{P}(\mathcal{M}_{\le \bar r} g(0) > \Lambda)$. Divide by $(2k)^2$, let $k \to \infty$, then $\bar r \to \infty$. Let $T_r f(y) = \int \varphi(z) f(y + rz)\,\mathrm{d}z$, and choose K_φ with $\operatorname{supp}\varphi \subset B(0, K_\varphi)$. Then $\sup_r |T_r f| \le A_\varphi \mathcal{M} f$ with $A_\varphi = \|\varphi\|_\infty |B(0, K_\varphi)|$. Splitting f at the level $\Lambda/(2A_\varphi)$ and applying (5.10) to $|f|^2 \mathbf{1}_{\{|f| > \Lambda/(2A_\varphi)\}}$ gives, for stationary f with $\mathbb{E}f(0)^2 < \infty$,

$$\mathbb{P}\Big(\sup_{r>0} |T_r f(0)| > \Lambda\Big) \le \frac{C}{\Lambda^2}\, \mathbb{E}\big[f(0)^2 \mathbf{1}_{\{|f(0)| > c\Lambda\}}\big] \qquad (\Lambda > 0)\,, \tag{5.11}$$

with $c = 1/(2A_\varphi)$. The truncation is what will survive the union bound over a fine mesh.

One window. Fix $0 < s < s' \le S$, $\eta > 0$ and $R \ge 2$, and put $\alpha = \varepsilon_L(s')$, $\beta = \varepsilon_L(s)$, $a = \alpha/R$, $b = R\beta$ and $f = D_a - D_b$, a stationary field carrying the frequencies between $1/b$ and $2/a$. Let $U_b(r)$ and $W_a(r)$ be the random variables with the spectral multipliers $\chi(b\xi)(\widehat{\varphi}(r\xi) - 1)$ and $(1 - \chi(a\xi))\widehat{\varphi}(r\xi)$. Then, for every width $r \in [\alpha, \beta]$,

$$D(\varphi_r) = T_r f(0) + D_b(0) + U_b(r) + W_a(r)\,. \tag{5.12}$$

The term $D_b(0)$ does not depend on r and cancels in oscillations over the window. Integrating the r-derivatives of the two remainders in $L^2(\mathbb{P})$ and summing over dyadic frequency shells gives

$$\mathbb{E} \sup_{\alpha \le r \le \beta} |U_b(r)|^2 \le C\Big(\frac{\beta}{b}\Big)^2 N(2/b)\,, \qquad \mathbb{E} \sup_{\alpha \le r \le \beta} |W_a(r)|^2 \le C R^{-2} V(a)\,, \tag{5.13}$$

and both right sides are at most $CR^{-2}SL$ for large L, since $\beta/b = R^{-1}$ and $V(a), V(b) \le SL + O(1)$. By Lemma 5.6 at the two ends of the window and the hypotheses of the proposition, $L^{-1/2} f(0) \Rightarrow \mathcal{N}(0, s' - s)$ and $L^{-1}\mathbb{E}f(0)^2 \to s' - s$. Weak convergence together with convergence of second moments gives uniform integrability of $L^{-1} f(0)^2$, hence convergence of the truncated second moments in (5.11). Each of the three r-dependent terms in (5.12) oscillates by at most twice its supremum, so if the oscillation of X_L^D over $[s, s']$ exceeds η, one of the three suprema exceeds $\eta\sqrt{L}/6$. Applying (5.11) with $\Lambda = \eta\sqrt{L}/6$ and Markov's inequality with (5.13), then letting $L \to \infty$ and afterwards $R \to \infty$, we obtain, with G a standard Gaussian variable,

$$\limsup_{L\to\infty} \mathbb{P}\Big(\sup_{s_1, s_2 \in [s, s']} |X_L^D(s_1) - X_L^D(s_2)| > \eta\Big) \le \frac{C(s' - s)}{\eta^2}\, \mathbb{E}\big[G^2 \mathbf{1}_{\{\sqrt{s'-s}\,|G| > c'\eta\}}\big]\,. \tag{5.14}$$

The order of limits matters: the window and R are fixed while $L \to \infty$, and R is sent to infinity before the windows are refined. The first window $[0, s']$ needs one change, because then $\beta = \mathrm{e}^{-1}$ and $b = R/\mathrm{e}$ need not be small: the first bound in (5.13) still holds and equals $CR^{-2}N(2\mathrm{e}/R)$, which is finite and independent of L; and $D_b(0)$ and $D(\varphi_{\mathrm{e}^{-1}})$ are fixed square-integrable variables, negligible after division by $\sqrt{L}$. Hence (5.14) also holds for $s = 0$.

Refining the windows. Divide $[0, S]$ into m windows of length S/m. An increment over a distance at most S/m lies in one window or in two adjacent ones, so a union bound and (5.14) at threshold $\eta/2$ give

$$\limsup_{L\to\infty} \mathbb{P}\Big(\sup_{|s-s'|\le S/m} |X_L^D(s) - X_L^D(s')| > \eta \Big) \le \frac{CS}{\eta^2}\, \mathbb{E}\big[G^2 \mathbf{1}_{\{|G|>c'\eta\sqrt{m/S}/2\}}\big] .$$

Without the truncation in (5.11), each window would contribute a bound of order $S/(m\eta^2)$, and the sum over the m windows would not be small. Here the right side tends to zero as $m \to \infty$, which proves Proposition 5.5.

5.4 Proof of Theorem C

The field $D = \mathscr{H}_t^\vartheta$ satisfies the hypotheses of Proposition 5.5. It is centred and square integrable by (2.2) and the remark before (5.2), and translation invariant by [BCC+26, Proposition 1.8]. With our normalisation of $\widehat{f}$, [BCC+26, (8.16) and Proposition 8.3] give (5.9) for $\mathscr{H}_1^\vartheta$, for every $\vartheta \in \mathbb{R}$. For general t, the scaling covariance (2.3) gives $\mathsf{C}_t^\vartheta(z) = \mathsf{C}_1^{\vartheta+\log t}(z/\sqrt{t})$, so the spectral measure of $\mathscr{H}_t^\vartheta$ is the image of that of $\mathscr{H}_1^{\vartheta+\log t}$ under $\xi \mapsto \xi/\sqrt{t}$, and its mass in $\bar{B}(0, R)$ is still asymptotic to $\log\log R$. By translation invariance we may take $x = 0$. The finite-dimensional hypothesis is Proposition 5.2, and the moment hypotheses are Proposition 5.1 and (5.2). Path continuity of X_L was noted after Proposition 5.5.

Let $L_k \to \infty$. Since $X_{L_k}(0) \to 0$ in probability, Proposition 5.5 and the Arzelà–Ascoli criterion show that the laws of X_{L_k} on $C([0, S])$ are tight. Coordinate evaluations are continuous on $C([0, S])$, so by Proposition 5.2 every subsequential limit has the finite-dimensional distributions of a standard Brownian motion on $[0, S]$, and finite-dimensional distributions determine a Borel probability measure on $C([0, S])$. Hence X_{L_k} converges in distribution to Wiener measure, and since the sequence L_k was arbitrary, so does X_L as $L \to \infty$.

Remark 5.7. The proof gives slightly more. For nonnegative $\varphi \in C_c^\infty(\mathbb{R}^2)$ of arbitrary integral, the limit is $(\int \varphi)B$; and finite-dimensional convergence holds with a different nonnegative test function of integral one at each of finitely many scale times, all coordinates sharing one Brownian motion, by (5.3). We do not consider several spatial points, rates of convergence, or couplings with a Brownian motion.

6 The Gaussian free field tangent

By the scaling covariance (2.3), the field at time t seen at scale ε is the field at the very long time $t\varepsilon^{-2}$, at the very negative parameter $\vartheta - 2\log(1/\varepsilon)$, seen at scale one. Write $\lambda = 2\log(1/\varepsilon) - \vartheta$, so that the parameter is $-\lambda$ and the time is $c\mathrm{e}^\lambda$ with $c = t\mathrm{e}^\vartheta$. As $\lambda \to \infty$ the disorder is weak on every fixed time interval, but the horizon $c\mathrm{e}^\lambda$ is exactly long enough for its accumulated effect to stay of order one; this is the critical scaling. After a time reversal, the field tested against f is the f-average of the logarithm of the flow started near each point at a small time and run to time

ce^{λ}. We cut this time interval at two places, a polynomial horizon $S_\lambda = \lambda^{O(1)}$ and a short time $n = \lambda^{-O(1)}$.

The long future. For a test function f of zero integral, the contribution of the time interval $[S_\lambda, ce^{\lambda}]$ is negligible after multiplication by $\sqrt{\lambda}$. Over this interval the polymer started near a point x has spread far beyond the support of f, so its contribution is nearly the same for all x in the support and is cancelled by $\int f = 0$. Making this quantitative requires a comparison of polymer endpoint profiles that keeps track of a common first-order correction (Lemma 6.3 and Proposition 6.4).

The short initial interval. Over $[n, S_\lambda]$ the disorder is weak: a flow started at time n from a heat kernel has mass with variance of order $\log\lambda/\lambda$, so its logarithm is close to its mass minus one, which is linear in the flow, and $\sqrt{\lambda/2}$ times these linear functionals converge to a Gaussian vector with covariance Q (Proposition 6.2). The interval $[0, n]$, where the disorder at small spatial scales is not weak, contributes little after spatial averaging, because the covariance of logarithms of the flow over that interval decays on the spatial scale $\sqrt{n}$ (Lemma 6.5 and Proposition 6.6).

Lemma 6.1, Lemma 6.5, Proposition 6.6, the proof of Proposition 6.7 and the assembly in Section 6.5 are complete, given the results they quote. Proposition 6.2, Lemma 6.3 and Proposition 6.4 adapt estimates of [BCC+26] and of the cited works, and we sketch their proofs.

6.1 Reduction and notation

Throughout this section $\lambda > 0$ is large and suppressed from the notation. For $0 < a < T$ and a real $f \in C_c^\infty(\mathbb{R}^2)$, put

$$\mathcal{L}_{a,T}(f) = \int_{\mathbb{R}^2} f(x) \log \mathscr{Z}_{a,T}^{-\lambda}\big(g_a^{(x)}, \mathbf{1}\big)\, \mathrm{d}x\,, \tag{6.1}$$

the f-average of the logarithm of the flow at parameter $-\lambda$ started at time a from a heat kernel of variance a and run to time T with flat terminal weight. We call the masses $\mathscr{Z}_{a,T}^{-\lambda}(g_a^{(x)}, \mathbf{1})$ heat-started masses; they have mean one. Their logarithms have an expectation that does not depend on x, so for $\int f = 0$ the variable $\mathcal{L}_{a,T}(f)$ is centred. For fixed λ and T, and for f with $\int f = 0$, $\mathcal{L}_{a,T}(f)$ converges in $L^2(\mathbb{P})$ as $a \downarrow 0$ to a limit $K_T(f)$ [BCC+26, (2.8), (2.11) and Theorem 5.1]; all these variables, for all horizons T, are defined on one stochastic heat flow.

Lemma 6.1. *Let $f_1, \dots, f_k$ be as in Theorem D, let $\varepsilon \in (0,1)$ be so small that $\lambda = 2\log(1/\varepsilon) - \vartheta > 0$, and put $c = te^{\vartheta}$. Then $(\mathscr{H}_t^{\vartheta}(f_{i,\varepsilon}^{(x)}))_{i\le k}$ has the law of $(K_{ce^{\lambda}}(f_i))_{i\le k}$.*

Proof. By translation invariance we may take $x = 0$, and $f_{i,\varepsilon}^{(0)} = \mathcal{S}_{\varepsilon^2} f_i$. The scaling covariance (2.3) with $\kappa = \varepsilon^2$ and time t/ε^2 gives $(\mathscr{H}_t^{\vartheta}(f_{i,\varepsilon}^{(0)}))_i \overset{\mathrm{d}}{=} (\mathscr{H}_{t\varepsilon^{-2}}^{\vartheta - 2\log(1/\varepsilon)}(f_i))_i$, and $\vartheta - 2\log(1/\varepsilon) = -\lambda$, $t\varepsilon^{-2} = ce^{\lambda}$. At a fixed time T, the flat field has the law of $(K_T(f_i))_i$: the regularisation (2.1) at time T has, by reversal of time in the flow, the law of the corresponding integrals of $\log \mathscr{Z}_{\delta,T}(g_\delta^{(x)}, \mathbf{1})$, and both converge to their limits in probability [BCC+26, (2.8), (2.11), (5.2)]. The centrings play no role since $\int f_i = 0$. □

Since $\sqrt{\lambda/2}/\sqrt{\log(1/\varepsilon)} \to 1$, Theorem D is equivalent to the statement that for every fixed $c > 0$,

$$\Big(\sqrt{\lambda/2}\, K_{ce^{\lambda}}(f_i)\Big)_{i\le k} \implies \mathcal{N}(0, \mathsf{Q}) \qquad (\lambda \to \infty)\,. \tag{6.2}$$

The fields K_T at different horizons T live on one flow, but this coupling is not the joint law of the KPZ field at different times; only the law at the single horizon ce^λ enters (6.2).

The second moment. The covariance of linear functionals of the flow is explicit. For real integrable f, f' with $\iint |f(y)f'(y')|\,|\log|y-y'||\,\mathrm{d}y\,\mathrm{d}y' < \infty$, $T>0$, and $P_s f = g_s * f$,

$$\operatorname{Cov}\big(\mathscr{Z}^{-\lambda}_{0,T}(f,\mathbf{1}),\mathscr{Z}^{-\lambda}_{0,T}(f',\mathbf{1})\big) = 4\pi\int_0^T \langle f, P_{2s}f'\rangle\, U_\lambda(T-s)\,\mathrm{d}s\,, \tag{6.3}$$

where $U_\lambda(u) = \int_0^\infty \mathrm{e}^{-(\lambda-\log u)z-\gamma z}\,\Gamma(1+z)^{-1}\,\mathrm{d}z$ and γ is the Euler–Mascheroni constant; signed f is understood through its positive and negative parts. This is the continuum limit of the first-collision decomposition [BCC+26, (4.2)] together with the renewal asymptotics of [BCC+26, Appendix A]; the same formula is [GT26, Section 2.1], see also [CSZ19]. Since $\mathrm{e}^{-\gamma z}/\Gamma(1+z)\le 1$, with equality to second order at $z=0$,

$$0 < U_\lambda(u) \le \frac{1}{\lambda-\log u} \qquad (u<\mathrm{e}^\lambda)\,, \qquad \lambda\,U_\lambda(u_\lambda)\to 1 \quad \text{whenever } \log u_\lambda = o(\lambda)\,. \tag{6.4}$$

If $\int f = \int f' = 0$, Lemma A.1 gives

$$4\pi\int_0^\infty \langle f, P_{2s}f'\rangle\,\mathrm{d}s = 2\mathsf{Q}(f,f')\,, \qquad \mathsf{Q}(f,f') = -\iint f(u)f'(u')\log|u-u'|\,\mathrm{d}u\,\mathrm{d}u'\,;$$

the factor 2 is the source of the normalisation $\sqrt{\lambda/2}$.

6.2 Linear functionals

The Gaussian limit in Theorem D comes from the following central limit theorem for linear functionals of the flow at a moderately long horizon.

Proposition 6.2. *Let $T_\lambda\to\infty$ with $\log T_\lambda = o(\lambda)$, let $\delta_\lambda\ge 0$ tend to zero (with P_0 the identity), and let $f_1,\dots,f_k\in C_c^\infty(\mathbb{R}^2)$ have zero integrals. Then $(\sqrt{\lambda/2}\,\mathscr{Z}^{-\lambda}_{0,T_\lambda}(P_{\delta_\lambda}f_i,\mathbf{1}))_{i\le k}\Rightarrow\mathcal{N}(0,\mathsf{Q})$.*

Sketch of proof. By (6.3)–(6.4) and dominated convergence, $\lambda\operatorname{Cov}(\mathscr{Z}^{-\lambda}_{0,T_\lambda}(f_i,\mathbf{1}),\mathscr{Z}^{-\lambda}_{0,T_\lambda}(f_j,\mathbf{1}))\to 2\mathsf{Q}(f_i,f_j)$; the integrand $\langle f_i,P_{2s}f_j\rangle$ is $O(s^{-2})$ at infinity because $\widehat{f_i}(0)=\widehat{f_j}(0)=0$. For fixed λ the mass $\mathscr{Z}^{-\lambda}_{0,T}(f,\mathbf{1})$ is a square-integrable martingale in T, by the flow property [CM26] and the independence of the flow over disjoint time intervals, so that for a fixed T_0

$$\lambda\,\mathbb{E}\big|\mathscr{Z}^{-\lambda}_{0,T_\lambda}(f,\mathbf{1})-\mathscr{Z}^{-\lambda}_{0,T_0}(f,\mathbf{1})\big|^2 \to 4\pi\int_{T_0}^\infty \|P_sf\|_2^2\,\mathrm{d}s \qquad (\lambda\to\infty)\,,$$

which is small when T_0 is large. At the fixed horizon T_0 we use the central limit theorem of Caravenna, Cottini and Rossi [CCR25, Theorem 1.2]. Their partition function averages over all starting sites of $\mathbb{Z}^2$ at spatial scale $\sqrt{N}$; it is the mean of two independent contributions, one from each parity class, and after the spatial change $y\mapsto\sqrt{2}\,y$ each has the law of $\mathscr{Z}^{-\lambda}_{0,T_0}(f,\mathbf{1})$ in our normalisation. Their limiting covariance [CCR25, (1.13)] becomes $2\pi\int_0^{T_0}\langle f,P_{2s}f'\rangle\,\mathrm{d}s$ for the mean. The centred contributions, multiplied by $\sqrt{\lambda}$, are tight by (6.3)–(6.4); along a convergent subsequence the two independent limits have a Gaussian sum, so each is Gaussian by Cramér's decomposition theorem, with twice the variance of the mean. Hence $\sqrt{\lambda}\,\mathscr{Z}^{-\lambda}_{0,T_0}(f,\mathbf{1})$ has a Gaussian limit with covariance $4\pi\int_0^{T_0}\langle f,P_{2s}f'\rangle\,\mathrm{d}s$. Letting first $\lambda\to\infty$ at fixed T_0 and then $T_0\to\infty$, the converging-together theorem and the Cramér–Wold device give the claim for $\delta_\lambda=0$. Finally, by (6.3), $\lambda\,\mathbb{E}|\mathscr{Z}^{-\lambda}_{0,T_\lambda}(P_{\delta_\lambda}f-f,\mathbf{1})|^2\le 8\pi\int_0^\infty\|P_s(P_{\delta_\lambda}f-f)\|_2^2\,\mathrm{d}s\to 0$. □

6.3 Localisation in time

We fix the times used below on a geometric grid. Let ε_ρ be the constant of [BCC+26, Proposition 7.4], which tends to zero as $\rho \to \infty$, and fix the ratio $\rho > 1$ so large that $\varpi = 2\varepsilon_\rho < \frac{1}{2}$. Put

$$J = \left\lceil \frac{2 \log \lambda}{|\log \varpi|} \right\rceil , \qquad S_\lambda = p_\lambda \rho^J , \tag{6.5}$$

where $p_\lambda \in [\lambda^2, \rho\lambda^2]$ is a point of the grid $ce^\lambda \rho^{-\mathbb{N}}$. Then $\varpi^J \le \lambda^{-2}$, $S_\lambda \le \lambda^d$ for a fixed $d = d(\rho)$, and S_λ depends only on λ, c and ρ. For a grid time $n \le S_\lambda$ we write $\mathsf{D} = \log_\rho(ce^\lambda/n)$ for the number of blocks between n and the horizon. Uniformly over these times, $\mathsf{D} \ge c_\rho \lambda$ for large λ, and $\mathsf{D} \asymp \lambda$ when $|\log n| = O(\log \lambda)$, as for all observation times used below.

The key estimate compares the masses, over the future after a grid time n, of two polymer endpoint profiles at time n. For a grid entrance $a' < n$, let $\mu^{(x)}_{a',n}$ be the law at time n of the polymer started at time a' from $g^{(x)}_{a'}$, that is, $\mathscr{Z}^{-\lambda}_{a',n}(g^{(x)}_{a'}, \cdot)$ normalised to mass one, and let $\mathsf{A}_T(a', x)$ be the mass from time n to time T started from $\mu^{(x)}_{a',n}$. By the flow property, $\mathsf{A}_T(a', x) = \mathscr{Z}^{-\lambda}_{a',T}(g^{(x)}_{a'}, \mathbf{1}) / \mathscr{Z}^{-\lambda}_{a',n}(g^{(x)}_{a'}, \mathbf{1})$.

Lemma 6.3. *There is $C < \infty$, depending only on c and ρ, such that for all large λ, all grid times $a \le m < n \le S_\lambda$ with $n/m = \rho^J$, and all $x \in \mathbb{R}^2$, there is an event G_x, measurable with respect to the disorder before time n, with*

$$\mathbb{P}(\mathsf{G}_x^c) \le C\mathsf{D}^{-1} , \qquad \mathbb{E}\big[\mathbf{1}_{\mathsf{G}_x} \big| \mathsf{A}_{ce^\lambda}(a, x) - \mathsf{A}_{ce^\lambda}(m, x) \big| \big] \le C\big(\mathsf{D}^{-3/2} + \varpi^J \mathsf{D}^{-1/2}\big) .$$

The point of the lemma is that the constant does not depend on the earlier entrance a, however small, and that the right side is $O(\lambda^{-3/2})$, while each profile differs from the heat profile by a correction of order $\mathsf{D}^{-1/2}$.

Sketch of proof. After scaling the horizon ce^λ to one, the parameter becomes $\log c$, fixed, and the block estimates of [BCC+26, Section 7] apply with constants that do not depend on λ. The block expansion of [BCC+26, (7.15)] writes the ratio $\mathscr{R}_{i,j}$ of the partition function over blocks $i, \dots, j$ to the product of the block masses as a sum, over families of disjoint intervals of blocks, of products of the normalised closed activities $\mathscr{C}_{u,v}$, denoted $\mathscr{D}^{\text{closed}}_{u,v}$ in [BCC+26, (7.16)]; empty ratios equal one. Selecting the first interval whose right endpoint is at least l gives the exact identity

$$\mathscr{R}_{i,j} - \mathscr{R}_{i,l-1} = \sum_{l_2=l}^{j} \sum_{l_1=i}^{l_2-1} \mathscr{R}_{i,\min(l_1,l)-1} \, \mathscr{C}_{l_1,l_2} \, \mathscr{R}_{l_2+1,j} , \qquad i \le l \le j ,$$

in which the three factors use disjoint blocks. With [BCC+26, (7.11), (7.21)] it shows that $\mathscr{R}_{i,j} - \mathscr{R}_{i,l-1}$ has L^2 norm $O((j - l + 1)/\mathsf{D})$ on the block event, independently of the number $j - i$ of earlier blocks. Let $\mu_{i,j}$ be the endpoint profile from blocks $i, \dots, j$, h_j the heat profile at the end of block j, $\mathscr{O}_{u,j}$ the normalised open activities of [BCC+26, (7.49)], and $\|\cdot\|_{\mathcal{H}}$ the seminorm of [BCC+26, (7.43)], the conditional second moment of the future mass of a signed profile of total mass zero. The expansion [BCC+26, (7.48)] gives

$$\mu_{i,j} - h_j - \sum_{u=i}^{j} \mathscr{O}_{u,j} = \frac{1}{\mathscr{R}_{i,j}} \sum_{u=i}^{j} \big(\mathscr{R}_{i,u-1} - \mathscr{R}_{i,j}\big) \mathscr{O}_{u,j} .$$

On the profile-good event $\mathscr{R}_{i,j}^{-1} \le 2$, and the Cauchy–Schwarz inequality, the ratio bound above and the bound $O(\varpi^{j-u}\mathsf{D}^{-1/2})$ of [BCC+26, (7.56)] on $\mathscr{O}_{u,j}$ bound the expected seminorm of the right side by $C\sum_u \frac{j-u+1}{\mathsf{D}}\varpi^{j-u}\mathsf{D}^{-1/2} \le C\mathsf{D}^{-3/2}$. The ratio differences and the open activities overlap in disorder, and no independence between them is used; the bound concerns the expected seminorm, not its square. This improves the bound D^{-1} of [BCC+26, (7.53)]. For the two profiles started at a and m, the first-order corrections from blocks after m coincide and cancel, those before m are $O(\varpi^{J}\mathsf{D}^{-1/2})$, and the expected seminorm of the difference bounds the expected difference of the future masses. □

Proposition 6.4. *For every $f \in C_c^\infty(\mathbb{R}^2)$ with $\int f = 0$, as $\lambda \to \infty$,*

$$\sqrt{\lambda}\,\big(K_{c\mathrm{e}^\lambda}(f) - K_{S_\lambda}(f)\big) \xrightarrow{\ \mathbb{P}\ } 0\,, \tag{6.6}$$

and $\sqrt{\lambda}\,(\mathcal{L}_{a,c\mathrm{e}^\lambda}(f) - \mathcal{L}_{a,S_\lambda}(f)) \to 0$ in probability uniformly over the grid entrances $a = p_\lambda\rho^{-j}$, $j \ge 0$.

Sketch of proof. By the flow property, $\mathcal{L}_{a,c\mathrm{e}^\lambda}(f) - \mathcal{L}_{a,S_\lambda}(f) = \int f(x)\log\mathsf{A}(x)\,\mathrm{d}x$, where $\mathsf{A}(x)$ is the mass from S_λ to $c\mathrm{e}^\lambda$ started from the endpoint profile at time S_λ of the polymer started at a. Let $\mathsf{B}(x)$ be the same quantity for the polymer started at the delayed time p_λ. Lemma 6.3, with the grid time $S_\lambda = p_\lambda\rho^J$ in the role of n, bounds $\mathbf{1}_{\mathsf{G}_x}|\mathsf{A}(x) - \mathsf{B}(x)|$ by $O(\lambda^{-3/2})$ in expectation. Write $\mathsf{B} = \mathsf{Z}_{c\mathrm{e}^\lambda}/\mathsf{Z}_{S_\lambda}$ with $\mathsf{Z}_T(x) = \mathscr{Z}^{-\lambda}_{p_\lambda,T}(g^{(x)}_{p_\lambda}, \mathbf{1})$.

Semiconvexity. For a stationary positive random measure μ and $\mathsf{F} = \log(g_s * \mu)$ with $\sigma^2 = \operatorname{Var}\mathsf{F}(0) < \infty$, the Hessian bound $\nabla^2\mathsf{F} \ge -s^{-1}I$ places each directional derivative between a forward and a backward difference quotient, up to $\pm h/(2s)$; taking L^2 norms, using stationarity and optimising in h gives $\|\partial_e\mathsf{F}(0)\|_{L^2(\mathbb{P})} \le 4\sigma^{1/2}s^{-1/2}$. Integrating along segments from 0 and using $\int f = 0$,

$$\Big\|\int f(x)\,\mathsf{F}(x)\,\mathrm{d}x\Big\|_{L^2(\mathbb{P})} \le \frac{4\sigma^{1/2}}{\sqrt{s}}\int |x||f(x)|\,\mathrm{d}x\,.$$

Logarithmic moments. Uniformly in x and in the entrance $a \le p_\lambda$, $\mathbb{P}(|\log\mathsf{A}(x)| > u) \le \min\{1, C\sqrt{\lambda}\,\mathrm{e}^{-u/3}\}$, and likewise for B. This follows the argument of [BCC+26, Lemma 7.8]: at an intermediate block at distance comparable to $\lambda\mathrm{e}^{u/3}$ from the horizon, the profile-good event fails with probability $O(\lambda^{-1}\mathrm{e}^{-u/3})$, the future mass has second moment $O(\lambda\mathrm{e}^{u/3})$ on it, and conditional Cauchy–Schwarz with [BCC+26, (3.5)] bounds the negative half moment of the remaining ratio; if that block precedes the entrance, the heat start itself is used. Hence $\|\log\mathsf{A}(x)\|_q + \|\log\mathsf{B}(x)\|_q \le C_q\log\lambda$.

A lower bound for B. Fix a cube E containing 0 and $\operatorname{supp} f$. By (6.3), $\mathbb{E}\mathsf{Z}_{c\mathrm{e}^\lambda}(0)^2 \le C\lambda$, so Gaussian convexity gives $\mathbb{E}\log\mathsf{Z}_{c\mathrm{e}^\lambda}(0) \ge -\frac12\log(C\lambda)$ and $\operatorname{Var}\log\mathsf{Z}_{c\mathrm{e}^\lambda}(0) \le \log(C\lambda)$. The derivative bound above gives $\nabla\log\mathsf{Z}_{c\mathrm{e}^\lambda}(0) = O_{\mathbb{P}}((\log\lambda)^{1/4}\lambda^{-1})$, and with $\nabla^2\log\mathsf{Z}_{c\mathrm{e}^\lambda} \ge -p_\lambda^{-1}I$ and Chebyshev's inequality, $\inf_E\log\mathsf{Z}_{c\mathrm{e}^\lambda} \ge -\frac{9}{16}\log\lambda$ with probability tending to one. Since $\operatorname{Var}\mathsf{Z}_{S_\lambda}(0) = O(\log\lambda/\lambda)$, $\log\mathsf{Z}_{S_\lambda} \to 0$ in probability at the vertices of E, and convexity of $\log\mathsf{Z}_{S_\lambda}(x) + |x|^2/(2p_\lambda)$ gives $\sup_E\log\mathsf{Z}_{S_\lambda} = o_{\mathbb{P}}(1)$. Hence $\mathbb{P}(\inf_E\mathsf{B} \ge \lambda^{-5/8}) \to 1$.

Conversion. On this event, and on $\mathsf{G}_x \cap \{|\mathsf{A}(x) - \mathsf{B}(x)| \le \lambda^{-5/8}/2\}$, the logarithm is Lipschitz with constant $2\lambda^{5/8}$, which contributes $O(\lambda^{1/2+5/8-3/2}) = O(\lambda^{-3/8})$ after multiplication by $\sqrt{\lambda}$. The rest of G_x has probability $O(\lambda^{-7/8})$ and G_x^c probability $O(\lambda^{-1})$; Hölder's inequality with exponent four and the logarithmic moments bound their contributions by $C_f\log\lambda\,(\lambda^{-5/32} + \lambda^{-1/4})$. By Fubini's theorem with $|f|$ and Markov's inequality, $\sqrt{\lambda}\int|f|\,|\log\mathsf{A} - \log\mathsf{B}| \to 0$ in probability, uniformly in a.

The delayed ratio. Apply the semiconvexity bound separately to $\log \mathsf{Z}_{c\mathrm{e}^\lambda}$ and $\log \mathsf{Z}_{S_\lambda}$, both with $s = p_\lambda \geq \lambda^2$ and variance $O(\log\lambda)$; since $\log \mathsf{B}$ is their difference, $\| \int f \log \mathsf{B}\|_{L^2(\mathbb{P})} \leq C_f(\log\lambda)^{1/4}\lambda^{-1}$, which is negligible after multiplication by $\sqrt{\lambda}$.

All bounds are uniform in j; letting $a \downarrow 0$ along the grid at fixed λ gives (6.6). □

6.4 The short initial interval

Proposition 6.4 reduces (6.2) to the field K_{S_λ} at the polynomial horizon. We next remove the interval $[0, n]$, where the disorder at small scales is not weak. The new ingredient is an estimate for positively quadrant dependent variables: the covariance of their logarithms is bounded by a polynomial in their moments times the square root of their covariance, so the spatial decay of the covariance of masses survives in their logarithms.

Lemma 6.5. *There is an absolute constant C with the following property. Let X, Y be positive random variables with $\mathbb{E}X = \mathbb{E}Y = 1$ such that $\mathbb{P}(X > u, Y > u') \geq \mathbb{P}(X > u)\mathbb{P}(Y > u')$ for all u, u'. Suppose that for some $V \geq 1$*

$$\mathbb{E}X^2,\ \mathbb{E}Y^2 \leq V\,, \qquad \mathbb{E}X^{-4},\ \mathbb{E}Y^{-4} \leq V^{10}\,, \qquad \mathbb{E}(\log X)^2,\ \mathbb{E}(\log Y)^2 \leq 4(1+\log V)^2\,.$$

Then $0 \leq \mathrm{Cov}(\log X, \log Y) \leq CV^{5/2}\sqrt{1+\log V}\,\sqrt{\mathrm{Cov}(X,Y)}$.

Proof. For increasing absolutely continuous functions Φ, Ψ with bounded derivatives,

$$\mathrm{Cov}\big(\Phi(X), \Psi(Y)\big) = \int_0^\infty \int_0^\infty \Phi'(u)\,\Psi'(u')\big[\mathbb{P}(X > u, Y > u') - \mathbb{P}(X > u)\mathbb{P}(Y > u')\big]\,\mathrm{d}u\,\mathrm{d}u'\,,$$

and the bracket is nonnegative. For $0 < \beta \leq 1$ let $\ell_\beta(z) = \log\max(z, \beta)$, which is increasing with $0 \leq \ell_\beta' \leq \beta^{-1}$, and $R_\beta(z) = \log z - \ell_\beta(z) = -(\log(\beta/z))_+$. The identity, applied with $\Phi = \Psi = \ell_\beta$ after truncation from above and passage to the limit in L^2, gives $0 \leq \mathrm{Cov}(\ell_\beta(X), \ell_\beta(Y)) \leq \beta^{-2}\,\mathrm{Cov}(X, Y)$, since the same integral without the derivatives equals $\mathrm{Cov}(X, Y)$. The discarded part is small: by Markov's inequality,

$$\mathbb{E}R_\beta(X)^2 = \int_0^\infty 2u\,\mathbb{P}\big(X < \beta\mathrm{e}^{-u}\big)\,\mathrm{d}u \leq \frac{\beta^4}{8}\,\mathbb{E}X^{-4} \leq \frac{\beta^4 V^{10}}{8}\,,$$

and the same for Y. Moreover $|\ell_\beta(z)| \leq |\log z|$ and $|R_\beta(z)| \leq |\log z|$. Writing $\log X = \ell_\beta(X) + R_\beta(X)$ and $\log Y = \ell_\beta(Y) + R_\beta(Y)$, each of the three covariances involving $R_\beta(X)$ or $R_\beta(Y)$ is at most $\|R_\beta(\cdot)\|_2$ times the L^2 norm of a logarithm, by the Cauchy–Schwarz inequality, so that

$$\mathrm{Cov}(\log X, \log Y) \leq \beta^{-2}\,\mathrm{Cov}(X, Y) + CV^5(1+\log V)\,\beta^2\,.$$

Since $\mathrm{Cov}(X, Y) \leq V - 1$, the choice $\beta^4 = \mathrm{Cov}(X, Y)/(CV^5(1+\log V))$ is admissible after enlarging C, and gives the upper bound; if $\mathrm{Cov}(X, Y) = 0$, let $\beta \downarrow 0$. The lower bound is the case $\beta \downarrow 0$ of the nonnegativity above. □

The heat-started masses satisfy the hypotheses of Lemma 6.5. They are limits in law of increasing functions of independent Gaussian disorder variables, so the Harris–FKG inequality on the lattice passes to their joint law through bounded monotone continuous functions. The moment hypotheses follow from Gaussian convexity: for a heat-started mass A,

$$\mathbb{E}A^{-4} \leq (\mathbb{E}A^2)^{10}\,, \qquad \mathrm{Var}(\log A) \leq \log \mathbb{E}A^2\,, \qquad -\mathbb{E}\log A \leq \tfrac{1}{2}\log \mathbb{E}A^2\,, \tag{6.7}$$

by [BCC+26, (3.2), (3.3), (3.5)] and their extension to the flow in [BCC+26, Section 3.4]. No smallness of the variance is needed.

Proposition 6.6. *There is an absolute constant C such that the following holds. Let $0 < a < n$ with $\lambda - \log n > 0$, put $\mathsf{v} = \log(n/a)/(\lambda - \log n)$ and $V = 1 + \mathsf{v}$, and let $A(x) = \mathscr{Z}^{-\lambda}_{a,n}(g_a^{(x)}, \mathbf{1})$. For every bounded integrable f,*

$$\operatorname{Var}\Big(\int_{\mathbb{R}^2} f(x) \log A(x)\, \mathrm{d}x\Big) \le C\|f\|_{L^1}\|f\|_{L^\infty}\, n\, V^3 \sqrt{1 + \log V}\,.$$

Proof. By the covariance formula (6.3) and the bound (6.4),

$$\operatorname{Cov}\big(A(x), A(y)\big) = \int_a^n \frac{\mathrm{e}^{-|x-y|^2/(4u)}}{u}\, U_\lambda(n-u)\, \mathrm{d}u \le \mathsf{v}\, \mathrm{e}^{-|x-y|^2/(4n)}\,;$$

in particular $\mathbb{E}A(0)^2 \le V$. Lemma 6.5 and (6.7) give

$$0 \le \operatorname{Cov}\big(\log A(x), \log A(y)\big) \le CV^{5/2}\sqrt{1+\log V}\,\sqrt{\mathsf{v}}\,\mathrm{e}^{-|x-y|^2/(8n)}\,.$$

Integrate against $|f(x)f(y)|$; the Gaussian factor integrates to $8\pi n$, and $\sqrt{\mathsf{v}} \le \sqrt{V}$. □

The variance bound grows like V^3, which is unbounded as $a \downarrow 0$ at fixed n, but it carries the factor n, the area over which the logarithms are correlated. If n is small, this factor absorbs a polynomial growth of V. This is how the proof combines an extremely fine smoothing scale with a short initial time interval. All times below lie on the grid $c\mathrm{e}^\lambda \rho^{-\mathbb{N}}$, and comparisons hold up to factors ρ. With J and S_λ as in (6.5), choose

$$a \asymp \exp(-\lambda^{40})\,, \qquad n \asymp \lambda^{-200}\,, \qquad m = n\rho^{-J}\,, \tag{6.8}$$

so that $a \ll m < n \ll 1 \ll S_\lambda \ll c\mathrm{e}^\lambda$. The exponents are not optimised; what matters is that $\lambda(\log(c\mathrm{e}^\lambda/a))^{-1/20} \to 0$ and $200 > 1 + 3 \cdot 39$.

Proposition 6.7. *For every $f \in C_c^\infty(\mathbb{R}^2)$ with $\int f = 0$, as $\lambda \to \infty$,*

$$\sqrt{\lambda}\,\Big(K_{S_\lambda}(f) - \mathscr{Z}^{-\lambda}_{m,S_\lambda}(P_m f, \mathbf{1})\Big) \xrightarrow{\ \mathbb{P}\ } 0\,. \tag{6.9}$$

Proof. Write $S = S_\lambda$.

Step 1: a fine entrance. The bounds of [BCC+26, Remarks 4.2 and 4.8] on the regularisation error keep track of the test function. After scaling the horizon to one, they bound the mean-square error for the test function ϕ and the smoothing δ by $C(\mathcal{E}_{\sqrt{\delta}}(\phi)/|\log\delta| + \|\phi\|_{L^1}^2/|\log\delta|^{1/20})$, also for test functions depending on δ, where

$$\mathcal{E}_\tau(\phi) = \iint |\phi(y)\phi(y')|\, 4\int_0^\tau g_{2s}(y-y')\, \mathrm{d}s\, \mathrm{d}y\, \mathrm{d}y'\,.$$

The same bounds hold for the limit as the second smoothing tends to zero, by the argument of [BCC+26, (2.17)], applied at fixed λ and fixed test function. Scaling the horizon $c\mathrm{e}^\lambda$ to one turns f into $\phi(z) = c\mathrm{e}^\lambda f(\sqrt{c\mathrm{e}^\lambda}\, z)$ and a into $\delta = a/(c\mathrm{e}^\lambda)$, with $\|\phi\|_{L^1} = \|f\|_{L^1}$ and

$$\mathcal{E}_{\sqrt{\delta}}(\phi) = \mathcal{E}_{\sqrt{ac\mathrm{e}^\lambda}}(f) \le 4\sqrt{ac\mathrm{e}^\lambda}\,\|f\|_{L^1}\|f\|_{L^\infty}\,.$$

Since $|\log\delta| \asymp \lambda^{40}$, it follows that $\lambda\,\mathbb{E}|K_{c\mathrm{e}^\lambda}(f) - \mathcal{L}_{a,c\mathrm{e}^\lambda}(f)|^2 \to 0$. Together with Proposition 6.4, applied to K and to $\mathcal{L}_{a,\cdot}$ at the grid entrance $a \le p_\lambda$, this gives $\sqrt{\lambda}\,(K_S(f) - \mathcal{L}_{a,S}(f)) \to 0$ in probability.

Step 2: the interval $[0, n]$. For the entrance a and the horizon n, $\mathsf{v} = O(\lambda^{39})$, and Proposition 6.6 gives

$$\lambda\,\mathbb{E}|\mathcal{L}_{a,n}(f)|^2 \le C_f \lambda\, n\, \lambda^{117}\sqrt{\log\lambda} = O\big(\lambda^{-82}\sqrt{\log\lambda}\big)\,;$$

the mean is zero since $\int f = 0$. For the entrance m, $\mathsf{v} = O(\log\lambda/\lambda)$, and similarly $\lambda\,\mathbb{E}|\mathcal{L}_{m,n}(f)|^2 \to 0$.

Step 3: comparing the futures. By the flow property,

$$\mathcal{L}_{a,S}(f) = \mathcal{L}_{a,n}(f) + \mathcal{L}_{m,S}(f) - \mathcal{L}_{m,n}(f) + \int f(x) \log \frac{\mathsf{A}_S(a,x)}{\mathsf{A}_S(m,x)}\,\mathrm{d}x\,, \tag{6.10}$$

with A_S as defined before Lemma 6.3. Lemma 6.3, with $\mathsf{D} \asymp \lambda$ and $\varpi^J \le \lambda^{-2}$, gives $\mathbb{P}(\mathsf{G}_x^c) \le C\lambda^{-1}$ and $\mathbb{E}[\mathbf{1}_{\mathsf{G}_x}|\mathsf{A}_{c\mathrm{e}^\lambda}(a,x) - \mathsf{A}_{c\mathrm{e}^\lambda}(m,x)|] \le C\lambda^{-3/2}$. The same bound holds with S in place of $c\mathrm{e}^\lambda$: conditionally on the disorder up to time S, the difference of the masses to $c\mathrm{e}^\lambda$ has conditional mean equal to the difference of the masses to S, and Jensen's inequality applies. We write $\mathsf{A} = \mathsf{A}_S(a,x)$ and $\mathsf{B} = \mathsf{A}_S(m,x)$. The masses started at a have second moments $O(\lambda^{39})$ by (6.3), so Gaussian convexity gives $\|\log\mathsf{A}\|_p + \|\log\mathsf{B}\|_p \le C_p\log\lambda$; only second and negative moments of these masses are used. For the masses started at m, $\mathsf{B} = \mathsf{Z}'_S/\mathsf{Z}'_n$ with $\mathsf{Z}'_T = \mathscr{Z}^{-\lambda}_{m,T}(g_m^{(x)}, \mathbf{1})$, and both have variance $O(\log\lambda/\lambda)$. Their lattice approximations satisfy the hypotheses of [CSZ25, Theorem 1.11] uniformly (bounded local-uniformity product, bounded localisation since $m \le T - m$, bounded variance), so, taking the lattice limit at each fixed λ first, $\mathbb{E}(\mathsf{Z}'_n)^{2p} \le C_p$; Gaussian convexity gives $\mathbb{E}(\mathsf{Z}'_S)^{-2p} \le C_p$, and the Cauchy–Schwarz inequality gives $\mathbb{E}\mathsf{B}^{-p} \le C_p$. Split according to G_x, $\{\mathsf{B} \ge \lambda^{-1/4}\}$ and $\{|\mathsf{A} - \mathsf{B}| \le \lambda^{-1/4}/2\}$. On the intersection the logarithm is Lipschitz with constant $2\lambda^{1/4}$, which contributes $O(\lambda^{1/2+1/4-3/2})$ after multiplication by $\sqrt{\lambda}$; the complements have probabilities $O(\lambda^{-5/4})$, $O(\lambda^{-1})$ and $O(\lambda^{-2})$, and Hölder's inequality with the logarithmic moments controls their contributions. Hence

$$\sqrt{\lambda}\,\mathbb{E}\int |f(x)|\,\big|\log\mathsf{A}_S(a,x) - \log\mathsf{A}_S(m,x)\big|\,\mathrm{d}x \to 0\,.$$

These events depend on x, and Fubini's theorem with $|f|$ is all that is used. With Steps 1 and 2 and (6.10), $\sqrt{\lambda}\,(K_S(f) - \mathcal{L}_{m,S}(f)) \to 0$ in probability.

Step 4: linearisation. For a heat-started mass Z with variance v, Gaussian convexity gives $\mathbb{E}|Z - 1 - \log Z| = -\mathbb{E}\log Z \le \frac{1}{2}\log(1+\mathsf{v})$. The masses $\mathscr{Z}^{-\lambda}_{m,S}(g_m^{(x)}, \mathbf{1})$ have variance $O(\log\lambda/\lambda)$, and $\int f(x)(\mathscr{Z}^{-\lambda}_{m,S}(g_m^{(x)}, \mathbf{1}) - 1)\,\mathrm{d}x = \mathscr{Z}^{-\lambda}_{m,S}(P_m f, \mathbf{1})$ because $\int f = 0$. Hence $\sqrt{\lambda}\,\|\mathcal{L}_{m,S}(f) - \mathscr{Z}^{-\lambda}_{m,S}(P_m f, \mathbf{1})\|_{L^1(\mathbb{P})} \le C\|f\|_{L^1}\log\lambda/\sqrt{\lambda} \to 0$, which together with Step 3 proves (6.9). □

6.5 Proof of Theorem D

The horizon S_λ and the times m of (6.8) do not depend on the test functions. By Propositions 6.4 and 6.7, applied to each f_i on the same flow, $\sqrt{\lambda}\,(K_{c\mathrm{e}^\lambda}(f_i) - \mathscr{Z}^{-\lambda}_{m,S_\lambda}(P_m f_i, \mathbf{1})) \to 0$ in probability for each i. The flow is invariant in law under translation in time [CSZ23, Theorem 1.2], so the vector $(\mathscr{Z}^{-\lambda}_{m,S_\lambda}(P_m f_i, \mathbf{1}))_i$ has the law of $(\mathscr{Z}^{-\lambda}_{0,S_\lambda - m}(P_m f_i, \mathbf{1}))_i$. Proposition 6.2, with $T_\lambda = S_\lambda - m$ and $\delta_\lambda = m$, shows that $\sqrt{\lambda/2}$ times this vector converges in law to $\mathcal{N}(0, \mathsf{Q})$. Slutsky's theorem gives (6.2), and Lemma 6.1 gives Theorem D.

Remark 6.8. The proof gives convergence of the finite-dimensional distributions only. It does not show that the covariances of the variables in Theorem D converge, nor tightness in a space of distributions, nor anything about the joint law of the field at several times; the fields K_T for different T are coupled through one flow in a way that does not reflect the time evolution of the KPZ field.

A The logarithmic kernel

Lemma A.1. *Let $f, f' \in C_c^\infty(\mathbb{R}^2)$ be real with $\int f = \int f' = 0$, write $P_s f = g_s * f$ and $\widehat{f}(\xi) = \int \mathrm{e}^{-i\xi\cdot y} f(y)\,\mathrm{d}y$. The integral $\mathsf{Q}(f,f') = -\iint f(u) f'(u') \log|u-u'|\,\mathrm{d}u\,\mathrm{d}u'$ converges absolutely, and*

$$\mathsf{Q}(f,f') = 2\pi \int_0^\infty \langle f, P_{2s} f'\rangle\,\mathrm{d}s = \frac{1}{2\pi}\int_{\mathbb{R}^2} \frac{\widehat{f}(\xi)\,\overline{\widehat{f'}(\xi)}}{|\xi|^2}\,\mathrm{d}\xi\,.$$

In particular, for $f_1, \dots, f_k$ as in Theorem D, the matrix $(\mathsf{Q}(f_i,f_j))_{i,j}$ is positive semidefinite, and $\mathcal{N}(0,\mathsf{Q})$ is the unique centred Gaussian law on $\mathbb{R}^k$ with this covariance.

Proof. The logarithm is locally integrable on $\mathbb{R}^2$ and f, f' are bounded with compact support, so the integral converges absolutely. For $T > 0$ and $z \neq 0$,

$$\int_0^T g_{2s}(z)\,\mathrm{d}s = \frac{1}{4\pi}\int_{|z|^2/(4T)}^\infty \frac{\mathrm{e}^{-w}}{w}\,\mathrm{d}w = \frac{1}{4\pi}\Big(-\gamma + \log(4T) - 2\log|z|\Big) + O\Big(\frac{|z|^2}{T}\Big)\,,$$

uniformly for z in compact sets. Integrating against $f(u)f'(u')$ with $z = u - u'$, the constant terms vanish because $\int f = 0$, and letting $T \to \infty$ gives the first identity. By Plancherel, $\langle f, P_{2s} f'\rangle = (2\pi)^{-2}\int \widehat{f}\,\overline{\widehat{f'}}\,\mathrm{e}^{-s|\xi|^2}\,\mathrm{d}\xi$, and integrating in s gives the second; the integrand is integrable near $\xi = 0$ because $\widehat{f}(0) = \widehat{f'}(0) = 0$. For $f = f'$ both expressions are nonnegative, which gives positive semidefiniteness, and a centred Gaussian law on $\mathbb{R}^k$ exists and is determined by any positive semidefinite covariance matrix, through its characteristic function. □

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